

पूर्व वैसाखि परीक्षा - २०८१

विषय सभ 'ख'

विषय : नेपाली (००११)

पुनर्दिष्ट : ३१

उत्तीर्णदिष्ट : १४

समय : १:३० मिनेट

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परीक्षार्थको सबैसम्बन्धी जानकारी प्राप्त गर्न विद्यार्थीहरूले आफ्नो सम्बन्धित विद्यालयको मुख्यालयमा पुगेर जानकारी लिनुपर्नेछ ।

विद्यार्थीहरूको उत्तरपत्र प्रत्यक्ष र शोधपत्रका आधारमा पहिचान गर्नुहोस् :

घ, च, ब

शुद्ध गरी पुनर्लेखन गर्नुहोस् :

जीवन मनुष्यको यस्तो गतिविधि हो, जसलाई हामीले बोधो, बुझो सम्झने र विचार गर्ने गर्दा रहेछ । यही यात्राको शिलशिलामा म दुर्गम भनिएको पहाड छाडेर काठमाडौं पुगे ।

अनुच्छेदमा रेखाङ्कन गरिएका शब्दको पर्याय पहिचान गरी लेख्नुहोस् :

रामका भाइले भलाई धेरै कुराहरू बोधे । उनले सोधेका कुराहरूमध्ये धेरै धेरै बुझिने ।

विद्यार्थीहरूलाई बहुवचनमा रूपान्तरण गर्नुहोस् :

तपाईं असाध्यै मिहिनेती हुनुहुन्छ । उसले हजुरको तार्किक गर्वबोधो । हजुरको मिहिनेतबाट म पनि प्रभावित भएको छु ।

विद्यार्थीहरूले अनुच्छेद पढी सोधिएका वस्तुगत प्रश्नको मिल्ने उत्तर दिनुहोस् :

जीवन मान्छे असल हुन सक्दैन । उसको मन, वचन र कर्म एकै गर्ने सम्भव छैन तर मान्छे असल जीवन मान्छे असल हुन सक्दैन । उसको मन, वचन र कर्म एकै गर्ने सम्भव छैन तर मान्छे असल जीवन मान्छे असल हुन सक्दैन ।

रुद्र र तिनका जोडीपिच्छेका मन, वचन र कर्ममा जति कम फरक भयो मान्छे उति गतिबोधो हुन्छ । त्यसका लागि ऊ स्पष्ट चिन्तक, स्पष्ट वक्ता र स्पष्ट कर्ता हुनुपर्छ । जसलाई सोचले छुटो पनि भन्छ । म साहित्यकारका मान्न पनि बाँचे, नाचे । साहित्यकर्मी, कलाकार हुनुपर्छ । जसलाई सोचले छुटो पनि भन्छ ।

जसलाई सोचले छुटो पनि भन्छ । जसलाई सोचले छुटो पनि भन्छ । जसलाई सोचले छुटो पनि भन्छ । जसलाई सोचले छुटो पनि भन्छ ।

जसलाई सोचले छुटो पनि भन्छ । जसलाई सोचले छुटो पनि भन्छ । जसलाई सोचले छुटो पनि भन्छ । जसलाई सोचले छुटो पनि भन्छ ।

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साहित्यकारको जीवन सम्बन्ध होइन । त्यो निस्वार्थ साहित्य सम्बन्ध हो । यो साहित्य सम्बन्ध हो, कलाको सम्बन्ध हो । त्यहाँ सह, दुःख, पैसा, पद, प्रतिष्ठा, शक्तिको कुनै सम्बन्ध छैन । साहित्यको सम्बन्ध हो । यो निस्वार्थ साहित्य सम्बन्ध हो । यो साहित्य सम्बन्ध हो, कलाको सम्बन्ध हो । त्यहाँ सह, दुःख, पैसा, पद, प्रतिष्ठा, शक्तिको कुनै सम्बन्ध छैन ।

प्रश्नहरू

क) लेखकले जीवनको कुन अवस्था साहित्यमा समर्पित गरे ?

अ) विचारबन्धन ब) युवावस्था द) बुढावस्था

ख) लेखकका सबैभन्दा नजिकका आत्मीय मित्र को हुन् ?

अ) विद्यालयका साथीहरू ब) कारोबारका कर्मचारीहरू

ग) लेखकले साहित्यकर्मिहरूका बिचको सम्बन्धलाई के नाम दिएका छन् ?

अ) सह र सहको सम्बन्ध ब) अज्ञान र ज्ञानको सम्बन्ध

घ) जीवन मान्छे भनेका को मान्छे हुन् ?

अ) प्राय मान्छे ब) सबै मान्छे द) केही मान्छे

५. विद्यार्थीहरूले कवितामा पढी सोधिएका प्रश्नको उत्तर दिनुहोस् :

पूर्व टिप्पणी दिनुपर्नेछ । सम्झौता जस्तो रङ्ग होला

पश्चिम राखी तिमी जस्तो रङ्ग होला

कर्मगृह र कर्मगृहको हुइला माटो बोल्छ होला

वीर पुर्खा ! तिमी गीत गइलाभरि गुन्छ होला ।

प्रश्नहरू

अ) वीर पुर्खाहरू नेपालको रचनाका लागि कहाँ कहाँ पुगेका थिए ?

आ) तिमी गीत गइलाभरि गुन्छ होला भन्नुको तात्पर्य के हो ?

७. 'वीर पुर्खा' कवितामा हावा वीर पुर्खाहरूको कस्तो वर्णन गरिएको छ ? 'वीर पुर्खा' कवितामा

आधारमा समीक्षात्मक उत्तर दिनुहोस् ।

८. विद्यार्थीहरूले कुनै एक शीर्षकमा २५० शब्दसम्ममा नघटाई निबन्ध लेख्नुहोस् ।

क) नेपालको छन विद्युतीय जर्जो

ख) विकासका लागि राजनीतिक स्थिरता

ग) मेरो राम्ररी यात्रा

सफलताको शुभकामना !!

एवम् सम्मान पनि प्राप्त गरिसकेको छन्

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

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समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

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समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

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त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

समस्त काठमाडौंमा आवास निर्माण गरिएको छ ।

त्यो निर्माणमा पुर्णतः समर्पित छिन् । यसै

उद्देश्यमा निरन्तर समर्पित भइरहने भएकाले

समस्त उपस्थितबाट समेत सम्मान गरिसकेको छ ।

1. The first step is to identify the problem or question that needs to be answered. This involves understanding the context and the specific information required.

2. *Значение* *глаголов*

[illegible][illegible]

в порядке деления

1. Penelitian adalah kegiatan yang dilakukan untuk mencari, menemukan, dan mengembangkan pengetahuan baru.
 2. Metode adalah cara atau prosedur yang digunakan untuk melakukan penelitian.
 3. Teori adalah kumpulan konsep yang saling berkaitan yang menjelaskan suatu fenomena.
 4. Model adalah representasi yang sederhana dari suatu sistem yang kompleks.
 5. Analisis adalah proses memecah suatu masalah menjadi bagian-bagian yang lebih kecil untuk memahami lebih dalam.
 6. Sintesis adalah proses menggabungkan bagian-bagian yang telah dianalisis untuk membentuk suatu keseluruhan yang baru.
 7. Evaluasi adalah proses menilai nilai atau kualitas dari suatu objek atau sistem.
 8. Validasi adalah proses memastikan bahwa suatu hasil penelitian atau model benar-benar sesuai dengan kenyataan.
 9. Verifikasi adalah proses memastikan bahwa suatu hasil penelitian atau model benar-benar sesuai dengan prosedur yang ditetapkan.
 10. Generalisasi adalah proses menarik kesimpulan umum dari hasil penelitian yang spesifik.

1. How many times did you visit the site?
2. How many times did you visit the site?
3. How many times did you visit the site?
4. How many times did you visit the site?
5. How many times did you visit the site?

1. **අනුමැතිය** ලබා දීම සඳහා අවශ්‍ය වන ප්‍රධාන කොටස්
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1. The following are the names of the persons who have been appointed to the various committees of the Board of Directors of the American Telephone and Telegraph Company, for the year ending December 31, 1911:

[illegible][illegible][illegible]

1. The first of these is the fact that the

14 : 2000000	(1000) 10000 : 2000	2000 000 : 2000
15 : 2000000	2000 000 : 2000	10 : 2000

KATHMANDU MODEL SECONDARY SCHOOL

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' MORNING [SET-B] F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

Multiple choice questions

1. The dimensions of Power are

- a. $[ML^2T^{-2}]$ b. $[M^{-1}L^2T^{-2}]$
c. $[M^{-1}L^2T^{-1}]$ d. $[M^{-1}L^2T^{-3}]$

2. Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?

- a. PQ-R b. PQ/R c. R/(P-Q) d. (PR-Q)/QR

3. Which of the following statements is false?

- a. Having magnitude and direction is sufficient to be a vector quantity
b. Distance is a scalar quantity but displacement is a vector quantity
c. Momentum, force and velocity are vector quantities
d. Mass, speed and energy are scalar quantities.

4. At what temperature do the Fahrenheit and Celsius thermometer shows the same reading?

- a. -40°F b. 212°F c. 32°F d. -81°F

5. Which of the following value of charge is not possible?

- a. $1.6 \times 10^{-18}\text{C}$ b. $3.2 \times 10^{-18}\text{C}$
c. $3.6 \times 10^{-20}\text{C}$ d. 0.5C

6. If a body is charged by rubbing, its mass

- a. Increases slightly b. Decreases slightly
c. remains same d. May increase or decrease slightly

7. When a glass rod is rubbed with woollen cloth

- a. Negative charges produced on wool, but glass rod remains neutral
b. equal and opposite charges are produced on the both

- c. equal and similar charges are produced on the both
d. positive charge is produced on the glass rod but no charge on wool.

GROUP 'B' [4x5=20]

Short answer questions

8. a. Name any two physical quantities which have same dimensions.
b. A body moving through air at high-speed V experiences a retarding force F given by $F = KAv^2$ where A is the surface area of body, ρ is the density of the air and K is numerical constant. Deduce the value of x.
c. Taking force, length and time to be fundamental quantities. Find the dimensional formula of velocity.
9. a. State parallelogram law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors.
b. The magnitude of two vectors is equal. Show that their resultant divides the angle between them.
10. a. Should a quantity having magnitude and direction be necessarily a vector? Explain with an example.
b. Can two vectors of same magnitude have resultant equal to either of them? Explain.
c. If $y = at + bt^2$, Where y is the distance and t is the time. What is the dimension and unit of c.
11. a. How can you charge a body positively by induction? Explain.
b. Explain the modern theory of electrification.

GROUP 'C' [2x8=8]

Long answer questions.

12. a. Define absolute zero temperature.
b. Among two thermometers, one with spherical bulb and another with cylindrical bulb, which will respond quickly to temperature change? Why?
c. A thermometer is not calibrated correctly. So it reads melting point of ice as -5°C . What is the temperature of boiling of water on the same scale if this thermometer reads 70°C instead of 60°C ?
d. Discuss thermal equilibrium as a working principle of mercury thermometer.

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE - TERM EXAMINATION 2081

Class: XI 'Science' MORNING [SET-A] F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

1. The dimensions of universal gravitational constant are

- a. $[M^{-1}L^2T^{-2}]$ b. $[M^{-1}L^2T^{-1}]$
c. $[M^{-1}L^2T^{-3}]$ d. $[M^{-1}L^2T^{-4}]$

2. Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?

- a. PQ-R b. PQ/R c. P-Q/R d. (PR-Q)/QR

3. Which of the following statements is false?

- a. A vector has only magnitude, but scalar has both magnitude and direction
b. Distance is a scalar quantity, but displacement is a vector quantity
c. Momentum, force and velocity are vector quantities
d. Mass, speed and energy are scalar quantities.

4. If change in temperature in Celsius scale is 45°C , the temperature change in Fahrenheit scale will be

- a. 45°F b. 212°F c. 32°F d. 81°F

5. Which of the following values is not possible?

- a. $1.6 \times 10^{-18}\text{C}$ b. $3.2 \times 10^{-18}\text{C}$
c. $3.6 \times 10^{-20}\text{C}$ d. $1.6 \times 10^{-17}\text{C}$

6. On charging by conduction mass of the body may

- a. increase b. decrease
c. remains same d. increase or decrease

7. When a glass rod is rubbed with silk

- a. Negative charges produced on silk but glass rod remains neutral
b. equal and opposite charges are produced on both
c. equal and similar charges are produced on both
d. positive charge is produced on the glass rod but no charge on silk.

GROUP 'B' [4x5=20]

Short answer questions

- a) Name a physical quantity which has unit but not dimension. [1]
b) A body moving through air at high-speed V experiences a retarding force F given by $F = KAv^2$ where A is the surface area of body, ρ is the density of the air and K is numerical constant. Deduce the value of x. [3]

dimensional formula of Power

9. a) State triangle law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors. [3]
b) What is the angle between a 2N force and 3N force so that their sum is 4N? [2]
10. a) Should a quantity having magnitude and direction be necessarily a vector? Explain with an example. [1]
b) Can two vectors of same magnitude have resultant equal to either of them? Explain. [2]
c) Show that work done, and energy have same dimensions. [2]
11. a) How can you charge a body positively by induction? Explain. [3]
b) Explain the meaning of conservation of charge. [1]
c) Iron chains touching to the ground are suspended from a vehicle carrying inflammable materials, why? [1]

GROUP 'C' [2x8=8]

Long answer question.

12. a) What is the temperature of a vacuum? [1]
b) What do you mean by thermal equilibrium? Why are object in contact come in thermal equilibrium? [2]
c) If a Fahrenheit thermometer reads 95° when the standard centigrade scale thermometer reads 33° , what is the error in the Fahrenheit thermometer? [3]
d) Explain the molecular concept of thermal energy. [2]

Best of luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' MORNING [SET-B] F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

Multiple choice questions

1. The dimensions of Power are

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GROUP 'B' [4x5=20]

Short answer questions

8. a. Name any two physical quantities which have same dimensions.
b. A body moving through air at high-speed V experiences a retarding force F given by $F = KAv^2$ where A is the surface area of body, ρ is the density of the air and K is numerical constant. Deduce the value of x.
c. Taking force, length and time to be fundamental quantities. Find the dimensional formula of velocity.
9. a. State parallelogram law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors.
b. The magnitude of two vectors is equal. Show that their resultant divides the angle between them.
10. a. Should a quantity having magnitude and direction be necessarily a vector? Explain with an example.
b. Can two vectors of same magnitude have resultant equal to either of them? Explain.
c. If $y = abx^{1/2}$, Where y is the distance and t is the time. What is the dimension and unit of c.
11. a. How can you charge a body positively by induction? Explain.
b. Explain the modern theory of electrification.

GROUP 'C' [1x8=8]

Long answer questions.

12. a. Define absolute zero temperature.
b. Among two thermometers, one with spherical bulb and another with cylindrical bulb, which will respond quickly to temperature change? Why?
c. A thermometer is not calibrated correctly. So it reads melting point of ice as -5°C . What is the temperature of boiling of water on the same scale if this thermometer reads 70°C instead of 60°C ?
d. Discuss thermal equilibrium as a working principle of mercury thermometer.

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' MORNING [SET-A] F.M: 35
Time: 1.5 hrs. Subject: Physics [1011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP 'A' MCQ [7x1=7]

1. The dimensions of universal gravitational constant are
a. $[M^{-1}L^2T^{-2}]$ b. $[M^{-1}L^2T^{-1}]$
c. $[M^{-1}L^2T^{-3}]$ d. $[M^{-1}L^2T^{-2}]$
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GROUP 'B' [4x5=20]

Short answer questions

- a) Name a physical quantity which has unit but not dimension. [1]
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b) Explain the meaning of conservation of charge. [1]
c) Iron chains touching to the ground are suspended from a vehicle carrying inflammable materials, why? [1]

GROUP 'C' [1x8=8]

Long answer question.

12. a) What is the temperature of a vacuum? [1]
b) What do you mean by thermal equilibrium? Why are object in contact come in thermal equilibrium? [2]
c) If a Fahrenheit thermometer reads 95° when the standard centigrade scale thermometer reads 33° , what is the error in the Fahrenheit thermometer? [3]
d) Explain the molecular concept of thermal energy. [2]

Best of luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift SET-A F.M.: 35
Time: 1:30 hrs. Subject: Mathematics (0071) P.M.: 14

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [5x1=5]

Rewrite the correct options of each question in your answer sheet.

- The statement $\sim (p \wedge q)$ is logically equivalent to
a) $\sim p \wedge q$ b) $\sim p \vee q$
c) $\sim p \vee \sim q$ d) $p \vee \sim q$
- The sentence $x:3=5$ is
a) True statement b) False statement
c) Compound statement d) Not a statement
- If the product of roots of the equation $mx^2 + (2m-1)x + 6x = 0$ is -2 then what is the value of m ?
a) 4 b) -3 c) $\frac{1}{4}$ d) $-\frac{1}{4}$
- Both roots of the equation $ax^2 + bx + c = 0$ are zero if
a) $b=0$ b) $c=0$
c) $b=0$ and $c=0$ d) $a=b$
- What is the value of $\lim_{x \rightarrow 0} \frac{x^2}{x^2}$?
a) 0 b) $\frac{1}{2}$ c) ∞ d) $-\infty$
- What is the value of $\lim_{x \rightarrow 3} \sqrt{x} = ?$
a) $\sqrt{5}$ b) 5 c) undefined d) $1/\sqrt{5}$
- What is the value of $\lim_{x \rightarrow \infty} \frac{x+4x^2}{x^2}$?
a) $\frac{1}{4}$ b) 4 c) 2 d) $\frac{1}{2}$

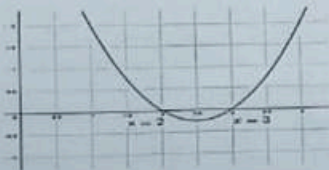
GROUP-B [5x5=25]

- Define Contradiction with example. [1]
- Write the negation of "If $5 < 6$ implies that $7 > 11$ then it is odd". [2]
- Construct truth table for following statement and comment the result
 $(p \vee q) \vee (\sim p)$. [2]
- Under what condition roots of quadratic equation are real and distinct? Also, explain it geometrically with example. [3]

- Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have negative roots. [1]
- If α and β are roots of $ax^2 + bx + c = 0$. Find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha^{-2} + \beta^{-2}$. [2]
- a. Define limit of a function at a point. [1]
b. Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$
(i) Using Table
(ii) Using Formula
(iii) Is it possible to find $\lim_{x \rightarrow 5} f(x)$? Explain. [3]

GROUP-C [10x1=10]

- Identify the hypothesis and conclusion for "Given that 2 is an even number then 5 is a prime number". [1]
- From following graph, [1]



- Find the quadratic equation. [1]
- If one root of a quadratic equation is $5 + \sqrt{3}$ then find the equation. [1]

d. Evaluate: [1]

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 + 2x - 8}$$

i) $\lim_{x \rightarrow 0} \frac{x^2 + 3x - 4}{x^2 - 2x + 1}$ [1]

ii) $\lim_{x \rightarrow 0} \frac{x^2 + 3x - 4}{x^2 - 2x + 1}$ [1]

iii) If $\lim_{x \rightarrow \infty} \frac{ax^2 + bx + c}{x^2 + 4x + 7} = \frac{11}{5}$ then find the value of a . [1]

The End

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift SET-A F.M.: 35
Time: 1.5 hrs. Subject: Physics (1011) P.M.: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A' MCQ [7x1=7]

Multiple choice questions

Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon the velocity v of water, its density ρ and the acceleration due to gravity g . Then m is directly proportional to
a. v^3 b. v^2 c. v d. v^4

Which of the following combinations of three dimensionally different physical quantities P, Q and R can never be a meaningful quantity?

- a. $PQ + R$ b. PQ/R c. $R/(P+Q)$ d. $(PQ^2)/QR$

Which of the following is the scalar quantity?

- a. Displacement b. Electrostatic force
c. Acceleration d. Work

At what temperature Celsius scales reading double the Fahrenheit reading?

- a. 12.30° b. -24.60° c. 24.60° d. -12.30°

How many electrons make up a charge of 4×10^{-5} C?

- a. 2.50×10^{14} b. 4.46×10^{14}
c. 2.50×10^6 d. 4.46×10^6

The soap bubble is given a positive charge, then its radius

- a. decreases
b. increases
c. remains unchanged
d. decreases or increases on the basis of magnitude of charge

A conducting wire is connected between two identical conducting spheres having a charge of -5C and +1 C respectively. Find out the new charge on each sphere.

- a. -4 C b. +4 C c. -2 C d. +2 C

GROUP-B' [4x5=20]

Answer question

- State principle of homogeneity. [1]
- Test dimensionally if the formula $t = 2\pi \sqrt{\frac{m}{F}}$ may be correct, where t is time period, m is mass, F is force and x is distance. [3]

dimensional formula of mass. [2]

- State triangle law of vector addition. Use this law to derive an expression for the magnitude and direction of the resultant of two vectors. [3]
- Two vectors have equal magnitudes, and their resultant also has the same magnitude. What is the angle between the two vectors? [2]
- Should a quantity having magnitude and direction be necessarily a vector? Explain with an example. [1]
- Can two vectors of same magnitude have resultant equal to either of them? Explain. [2]
- If $y = a + bt + ct^2$, where y is the distance and t is the time. What is the dimension and unit of c . [2]
- How can you charge a body negatively by induction? Explain. [3]
- Explain the meaning of the quantization of charge. [1]
- Does charge by friction follow conservation of charge? [1]

GROUP-C' [1x8=8]

Long answer questions.

- Define temperature on the basis of zeroth law of thermodynamics. [1]
- If two bodies are in thermal equilibrium in one frame, will they be in thermal equilibrium in all frames? Explain. [2]
- The steam point and ice point of a mercury thermometer are marked as 80° and 20° respectively. What will be the temperature in centigrade scale when this thermometer reads 52° ? [3]
- Write a short note about the absolute scale of temperature. [2]

Best of Luck



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Class: XI 'Science'

Time: 1:30 hrs.

MORNING SHIFT 'SET-A'

Subject: Chemistry [3011]

F.M: 37

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS

- $B_2S + H_2O_2 \rightarrow S + H_2O$ shows
 - Reducing action of H_2O_2
 - Oxidizing action of H_2O_2
 - Catalytic action of H_2O_2
 - Oxidizing action of H_2S
- Strongest oxidizing agent among the following is
 - $HClO$
 - $HClO_2$
 - $HClO_3$
 - $HClO_4$
- Oxidation state and valency of carbon in CH_2Cl_2 is
 - 0, 4
 - 0, 2
 - +4, 0
 - 4, 0
- Absolute zero is the temperature equal to
 - Room temperature
 - $273^\circ C$
 - $-273.15^\circ C$
 - $273.15^\circ C$
- The curve plotted between P and V at constant temperature is called
 - Isotherm
 - Isobar
 - Isochore
 - Isotope
- Which non-metal is most active?
 - F_2
 - N_2
 - O_2
 - S
- Which of the following species has the highest electron affinity?
 - F^-
 - O
 - O^-
 - Na^+
- The size of following species increase in the order
 - $Mg^{+2} < Na^+ < F^- < Al^{+3}$
 - $F^- < Al < Na^+ < Mg^{+2}$
 - $Al < Mg^{+2} < F^- < Na^+$
 - $Na^+ < Al < F^- < Mg^{+2}$
- The density of NH_3 at $30^\circ C$ and 5 atm pressure is
 - 2.30 gm/lit
 - 3.42 gm/lit
 - 23 gm/lit
 - 30 gm/lit

GROUP - 'B' [4×5=20]

SHORT ANSWER QUESTIONS

- State Modern Periodic Law. [1]
 - Write the main features of Modern Periodic Table. [4]
- Define the terms:
 - Screening Effect
 - Effective Nuclear Charge
 [1+1]
 - Define ionisation energy. Why is the ionisation energy of nitrogen higher than that of Oxygen? [1+2]
- Derive combined gas equation. [2]
 - Find the value of R in C.G.S. unit. [2]
 - What is significance of R? [1]
- Volume of moist gas at $23^\circ C$ and 800 mm Hg pressure is 100 cc. Calculate the aqueous tension at $23^\circ C$ if the volume occupied by gas at NTP is 94.53 cc. [3]
 - Write any two applications of Boyle's Law. [2]

GROUP - 'C' [1×8=8]

LONG ANSWER QUESTION

- Differentiate between oxidation number and valency. [2]
 - Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent in the reaction,

$$KMnO_4 + KCl + H_2SO_4 \rightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$$
 [2]
 - Justify that, " SO_2 acts as both oxidizing agent as well as reducing agent". [2]
 - Calculate the oxidation number of underlined species.
 - $N(CO)_x$
 - $Cr(NH_4)_2(SO_4)_3$
 - NH_4NO_3
 - SO_4^{2-}
 [2]

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' (MORNING) SET-A F.M: 35
Time: 1:30 hrs. Subject: Chemistry [3011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

- Which of the following species has the highest electron affinity?
(a) F^- (b) O (c) O^+ (d) Na^+
- Pick out the isoelectronic structures from the following.
(i) CH_4 (ii) H_2O^+ (iii) NH_3 (iv) CH_3
(v) SiH_4 (vi) CO (vii) NO (viii) CO_2
- The gas law giving relationship between volume and temperature of a gas is
(a) Boyle's law (b) Charles's law
(c) Graham's law (d) Dalton's law
- Oxidation number of phosphorus ranges from
(a) -3 to +5 (b) +3 to -5 (c) -2 to +6 (d) +2 to -6
- The reaction, $H_2S + H_2O_2 \rightarrow S + H_2O$ shows,
(a) Reducing action of H_2O_2 (b) Oxidizing action of H_2O_2
(c) Catalytic action of H_2O_2 (d) Reducing action of H_2S

GROUP 'B' [5x1=5]

- Short Questions.
- Show with examples that all reactions are not redox reactions. [2]
 - Distinguish between valency and oxidation number. [2]
 - Identify the oxidation number of S in H_2SO_5 . [1]
 - Balance the following redox reaction by oxidation number method.
 $K_2Cr_2O_7 + FeSO_4 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + H_2O$ [1]
 - Identify Oxidizing and reducing agent in above reaction. [1]
 - Define absolute zero temperature according to Charles's law with the help of suitable graph. [2]
 - Derive the ideal gas equation. [1]
 - What is the application of Boyle's law? [1]
 - An open vessel at $27^\circ C$ is heated until three-fifths of the air in it has been expelled. Assuming that the volume of vessel remains constant, find the temperature to which the vessel has been heated? [3]
 - Plot the graph for
(i) Volume Vs 1/pressure (ii) Pressure X volume Vs Pressure [1+1]
 - Define the terms (i) Screening Effect (ii) Effective Nuclear Charge [1+1]
 - Define electronegativity. Why is the electronegativity of chlorine greater than that of fluorine? [1+2]
 - State modern periodic law. Write the advantages of modern periodic table. [1+4]

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' (MORNING) SET-B F.M: 35
Time: 1:30 hrs. Subject: Chemistry [3011] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

- The process requiring the absorption of energy is
(a) $F \rightarrow F^-$ (b) $Cl \rightarrow Cl^-$ (c) $O \rightarrow O^{2-}$ (d) $H \rightarrow H^-$
- Which has the highest electronegativity?
(a) C (b) Mg (c) O (d) S
- Oxidation state of oxygen atom in potassium superoxide is
(a) $-\frac{1}{2}$ (b) -1 (c) -2 (d) 0
- A solution of SO_2 in water reacts with H_2S precipitating sulphur. Here, SO_2 acts as
(a) an oxidizing agent (b) a reducing agent
(c) an acid (d) a catalyst
- A gas is initially at 1 atm pressure. To compress it to $\frac{1}{8}$ th of its initial volume, pressure to be applied is
(a) 1 atm (b) 2 atm (c) 4 atm (d) $\frac{1}{8}$ atm

GROUP 'B' [5x1=5]

- Calculate the oxidation number of Cr in CrO_5 with structure. [2]
- HNO_3 acts as oxidizing and agent, why? [2]
- Identify oxidation and reduction reaction as well as oxidant and reductant in the given reaction.
 $K_2Cr_2O_7 + FeSO_4 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + H_2O$
- Balance the following redox reaction by oxidation number method.
 $Br^- + MnO_4^- + H_2SO_4 \rightarrow KHSO_4 + MnSO_4 + Br_2 + H_2O$
- Give example for maximum and minimum oxidation number of Cl. [2]
- Derive combined gas equation. [2]
- Find the value of 'n' in SI unit. [1]
- Give the application of Charles's law. [1]
- A balloon can hold 1000 cc of air before bursting. The balloon can hold 975 cc of air at $5^\circ C$. Will it burst when it is taken into a room at $25^\circ C$? Assume that the pressure of the gas in the balloon remains constant.
(i) Explain the nature of curve obtained during isothermal conditions.
(ii) Compare the ionic size of (i) Na and Na^+ (b) Cl and Cl^- [1+1]
- Explain the terms (i) Ionization potential. What are the various factors that affect ionization potential? Explain. [1+2]
- Why is the ionization potential of Be higher than that of B? [2]

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift, Set-B F.M: 35
Time: 1:30 hrs. Subject: Mathematics [0671] P.M: 14

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

Write the correct options of each question in your answer sheet.

- The statement $\sim (p \vee q)$ is logically equivalent to
(a) $\sim p \wedge \sim q$ (b) $\sim p \wedge q$
(c) $p \vee \sim q$ (d) $p \vee q$
- The value of $\lim_{x \rightarrow 0} \frac{x^2 + 10x}{x + 4}$ is
(a) $\frac{1}{2}$ (b) 2 (c) 4 (d) 8
- The sentence $x = 2$ is
(a) True statement (b) False statement
(c) Compound statement (d) Not a statement
- Both roots of the equation $x^2 + bx + c = 0$ are zero if
(a) $b = 0$ (b) $c = 0$
(c) $b = 0$ and $c = 0$ (d) abc
- The value of $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1}$ is
(a) 0 (b) $\frac{1}{2}$ (c) ∞ (d) $-\infty$
- The value of $\lim_{x \rightarrow 0} \log x$ is
(a) 12 (b) 0 (c) undefined (d) 5
- If the product of roots of the equation $mx^2 + (2m+1)x - 6 = 0$ is -2 then what is the value of m?
(a) 3 (b) -3 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

GROUP 'B' [5x1=5]

- Define contradiction with example. [1]
- Write the negation of "If $5 < 6$ implies that $7 > 11$ then 8 is odd". [2]
- Construct truth table for following statement and comment on the result $(p \vee q) \vee (\sim p)$. [3]
- Under what condition roots of the quadratic equation are real and distinct? Also, explain it geometrically with an example. [3]

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Day Shift, Set-B F.M: 35
Time: 1:30 hrs. Subject: Mathematics [0671] P.M: 14

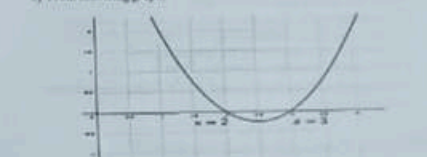
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [5x1=5]

Write the correct options of each question in your answer sheet.

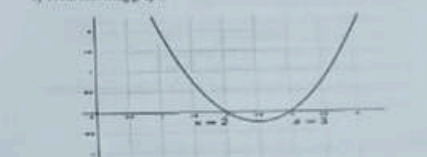
- The statement $\sim (p \vee q)$ is logically equivalent to
(a) $\sim p \wedge \sim q$ (b) $\sim p \wedge q$
(c) $p \vee \sim q$ (d) $p \vee q$
- The value of $\lim_{x \rightarrow 0} \frac{x^2 + 10x}{x + 4}$ is
(a) $\frac{1}{2}$ (b) 2 (c) 4 (d) 8
- The sentence $x = 2$ is
(a) True statement (b) False statement
(c) Compound statement (d) Not a statement
- Both roots of the equation $x^2 + bx + c = 0$ are zero if
(a) $b = 0$ (b) $c = 0$
(c) $b = 0$ and $c = 0$ (d) abc
- The value of $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1}$ is
(a) 0 (b) $\frac{1}{2}$ (c) ∞ (d) $-\infty$
- The value of $\lim_{x \rightarrow 0} \log x$ is
(a) 12 (b) 0 (c) undefined (d) 5
- If the product of roots of the equation $mx^2 + (2m+1)x - 6 = 0$ is -2 then what is the value of m?
(a) 3 (b) -3 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$



- Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have negative roots. [3]
- If α and β are roots of $ax^2 + bx + c = 0$, find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha^{-1} + \beta^{-1}$. [3]
- Define limit of a function $f(x)$ at a point a . [1]
- Evaluate $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1}$. [2]
- Using Table. [2]
- Using Formula. [2]
- Is it possible to find $f(1)$? Explain. [1]

GROUP 'C' [5x1=5]

- Evaluate $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1}$. [2]
- Evaluate $\lim_{x \rightarrow 0} \frac{x^2 + 10x}{x + 4}$. [1]
- If $\lim_{x \rightarrow 0} \frac{x^2 + 10x}{x + 4} = \frac{10}{5}$ then find the value of a . [1]
- Identify the hypothesis and conclusion for "Given that 2 is an even number then 3 is a prime number". [1]
- From following graph.



- Find the quadratic equation. [2]
- If one root of a quadratic equation is $5 - \sqrt{3}$, find the equation. [3]

"The End"



KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2082

Class: XI 'Science'

Time: 1:30 hrs.

MORNING SHIFT 'SET-B'

Subject: Chemistry [3011]

F.M: 37

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS

- The oxide which cannot act as reducing agent is
a) NO_2 b) CO_2 c) SO_2 d) ClO_2
- Oxygen has positive oxidation state in
a) H_2O_2 b) Cl_2O c) F_2O d) NaOCl
- Oxidation number of two chlorine atoms in bleaching powder (CaOCl_2) are
a) -1, -1 b) +1, +1 c) +1, -1 d) 0, -1
- If both temperature and volume of a gas are doubled, the pressure
a) is doubled b) remains unchanged
c) is reduced to half d) is tripled
- One litre of an unknown gas weighs 1.25 gm at NTP. Which of the following gas corresponds above data?
a) NO_2 b) N_2 c) CO_2 d) O_2
- The outermost electronic configuration of the most electronegative element is
a) ns^2np^3 b) ns^2np^4 c) ns^2np^5 d) ns^2np^6
- Correct order of electron affinities is
a) $\text{Cl} > \text{Br} > \text{I} > \text{F}$ b) $\text{Cl} > \text{F} > \text{Br} < \text{I}$
c) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ d) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
- Pick the isoelectronic structures from the following
i) CH_3^+ ii) H_2O^+ iii) NH_3 iv) CH_3^-
a) (i) & (ii) b) (iii) & (iv) c) (i) & (iii) d) (ii), (iii) & (iv)
- A gas which perfectly obeys gas laws at all temperature is called
a) Noble gas b) permanent gas c) Inert gas d) Ideal gas

GROUP - 'B' [4×5=20]

SHORT ANSWER QUESTIONS

- a) Define periodicity. [1]
b) Describe Bohr's classification of elements. [4]
- a) Define electron affinity. [1]
b) Compare the size of Na^+ , Mg^{++} & Al^{+++} with reasons. [2]
c) Why is electron affinity of chlorine higher than fluorine? [2]
- a) State Boyle's law and hence derive its expression. [1+1]
b) Graphically represent all the isotherms. [3]
- a) Define absolute zero with graph. [1+1]
b) A flask of 2.5 litres capacity can withstand a pressure of 14 atm and is filled with CO_2 gas at 27°C and 760 mm Hg pressure. At what temperature will it explode on heating? [3]

GROUP - 'C' [1×8=8]

LONG ANSWER QUESTION

- a) Justify the statement, "Oxidation and reduction occurs simultaneously". [2]
b) Define autoredox reaction with example. [2]
c) Calculate the oxidation number of underlined species.
i) $\text{K}_4[\text{Fe}(\text{CN})_6]$ ii) $\text{C}_6\text{H}_{12}\text{O}_6$
iii) H_2HNO_3 iv) Fe_2O_4 [2]
d) Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent. [2]
 $\text{K}_2\text{Cr}_2\text{O}_7 + \text{FeSO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Morning Shift Set-A P.M.: 35
Time: 1:30 hrs. Subject: Mathematics (0071) P.M.: 14
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [5x1=5]

Rewrite the correct options of each question in your answer sheet.

- What is the contrapositive of $p \Rightarrow q$?
a) $p \Rightarrow \neg q$ b) $q \Rightarrow p$ c) $\neg q \Rightarrow \neg p$ d) $p \Rightarrow q$
- What is the negation of $p \Rightarrow q$?
a) $p \vee q$ b) $\neg p \wedge \neg q$ c) $p \wedge q$ d) $\neg p \vee q$
- If the roots of the equation $ax^2 + bx + c = 0$ are equal in magnitude but opposite in sign then
a) $a = 0$ b) $b = 0$ c) $c = 0$ d) $a \neq b$
- If α and β are roots of $ax^2 + bx + c = 0$, what is the value of $\alpha\beta^2 + \alpha^2\beta - \alpha\beta$?
a) $\frac{c(a-b)}{a^2}$ b) $\frac{(a-b)}{a^2}$ c) $-\frac{bc}{a^2}$ d) $\frac{(a+b)}{a^2}$
- What is the value of $\lim_{x \rightarrow 0} \frac{1}{x}$ is
a) 0 b) $\frac{1}{0}$ c) ∞ d) $-\infty$
- What is the value of $\lim_{x \rightarrow 0} \pi$?
a) 5 b) 8 c) undefined d) π
- What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+x^2}}{x}$?
a) $\frac{1}{x}$ b) 4 c) $\frac{1}{2}$ d) $\frac{1}{x}$

GROUP-B [5x5=25]

- Define tautology with an example. [1]
- Write the negation of "If $3 < 4$ implies that $3 > 7$ then 4 is odd". [2]
- Construct truth table for following statement and comment the result
($p \wedge q$) $\Rightarrow p \vee q$. [2]
- Under what condition the roots of quadratic equation are real and equal? Also, interpret it geometrically. [3]
- Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have positive roots. [2]
- If α and β are roots of $ax^2 + bx + c = 0$. Find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha^{-2} + \beta^{-2}$. [5]

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2081

Class: XI 'Science' Morning Shift Set-B P.M.: 35
Time: 1:30 hrs. Subject: Mathematics (0071) P.M.: 14
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [5x1=5]

Rewrite the correct options of each question in your answer sheet.

- What is the contrapositive of $q \Rightarrow p$?
a) $p \Rightarrow \neg q$ b) $q \Rightarrow p$ c) $q \Rightarrow \neg p$ d) $p \Rightarrow q$
- What is the negation of $q \Rightarrow p$?
a) $p \vee q$ b) $p \wedge \neg q$ c) $q \wedge \neg p$ d) $\neg p \vee q$
- If the roots of the equation $ax^2 + bx + c = 0$ are equal in magnitude but opposite in sign then
a) $a = 0$ b) $b = 0$ c) $c = 0$ d) none
- If α and β are roots of $ax^2 + bx + c = 0$, what is the value of $\alpha\beta^2 + \alpha^2\beta - \alpha\beta$?
a) $\frac{-c(a-b)}{a^2}$ b) $\frac{(a-b)}{a^2}$ c) $-\frac{bc}{a^2}$ d) $\frac{-(a+b)}{a^2}$
- What is the value of $\lim_{x \rightarrow 0} \frac{1}{x^2}$ is
a) 0 b) $\frac{1}{0}$ c) ∞ d) $-\infty$
- What is the value of $\lim_{x \rightarrow 0} 10$?
a) 5 b) 8 c) undefined d) 10
- What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+25x^2}}{3+5x}$?
a) $\frac{1}{3}$ b) 5 c) 2 d) 1

GROUP-B [5x5=25]

- Write the truth value of "3 is odd and 4 is even". Also, write its negation. [1]
- Construct truth table for the following statement and comment on its result
($p \wedge \neg q$) $\wedge$ ($p \Rightarrow q$) [2]
- Under what condition roots of the quadratic equation are imaginary? Also, interpret it geometrically. [3]
- Find the condition for quadratic equation $ax^2 + bx + c = 0$ to have positive roots. [2]
- If α and β are roots of $ax^2 + bx + c = 0$, find the quadratic equation whose roots are $\alpha^2 + \beta^2$, $\alpha^{-2} + \beta^{-2}$. [5]

a) Define limit of a function $f(x)$ at a point $x=a$. [1]

b) Evaluate $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ [1]

c) Using Table [2]

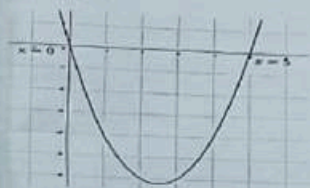
d) Using Formula [1]

iii) Is it possible to find $f(2)$? explain. [1]

GROUP-C [8x1=8]

a) Identify the hypothesis and conclusion for "If 8 is multiple of 2 then 8 is even". [1]

b) From following graph,



Find the quadratic equation. [2]

c) If one root of a quadratic equation is $2 + i\sqrt{3}$ then find the equation. [1]

d) Evaluate:

$$i) \lim_{x \rightarrow 0} \frac{x^2-3}{x^2+3x-12}$$
 [2]

$$ii) \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2}-1}{x^2}$$
 [1]

$$iii) \lim_{x \rightarrow \infty} \frac{3x^2+2x}{x^2-5x+7} = \frac{12}{5}$$
 then find the value of p . [1]

"The End"

11. a) Define limit of a function $f(x)$ at a point $x=a$. [1]

b. Evaluate $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ [1]

(i) Using Table [2]

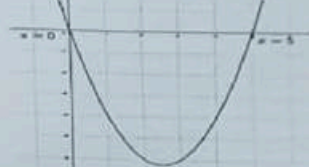
(ii) Using Formula [1]

(iii) Is it possible to find $f(2)$? explain. [1]

GROUP-C [8x1=8]

12. a) Identify the hypothesis and conclusion for "If a figure is parallelogram, its diagonal bisect each other". [1]

b) From following graph,



Find the quadratic equation. [2]

c) If one root of a quadratic equation is $2 + i\sqrt{3}$, find the equation. [1]

d) Evaluate:

$$i) \lim_{x \rightarrow 0} \frac{x^2-3}{x^2+3x-12}$$
 [2]

$$ii) \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2}-1}{x^2}$$
 [1]

$$iii) \lim_{x \rightarrow \infty} \frac{3x^2+2x}{x^2-5x+7} = \frac{12}{5}$$
 then find the value of p . [1]

"The End"

6. Choose the best options:

- a) The prefix 'dis-' goes with the word [5x1=5]
i) obey ii) possible iii) legal iv) happy
- b) The meaning of the prefix in the word 'preview' is
i) again ii) before iii) wrong iv) not
- c) The mountain is very **steep**. Choose the correct meaning of the word 'steep'.
i) flat ii) sloppy iii) wide iv) low
- d) If the guide words are "king – kite", which of the following will appear on that page?
i) kind ii) kitten iii) kitchen iv) keep
- e) 'Glory' comes before 'Gold' because comes before in the alphabet. (Fill in the blanks with suitable letters)
i) l/o ii) d/r iii) r/y iv) g/o

[Best Wishes]

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION- 2082

Class: XI
 Time: 1hr.

Morning Shift Set 'A'
 Subject: Computer Science [4271]

F. M.: 25
 P. M.: 10

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP – 'A' MCQ [5×1=5]

Multiple Choice Questions.

- The brain of Computer System is?
 a) ALU b) Memory c) Control Unit d) CPU
- Microprocessor was introduced in which generation of computer?
 a) Third b) Second c) Fourth d) First
- Which of the following is exclusively a sequential access storage device?
 a) Hard Disk b) Floppy Disk c) DVD d) Magnetic Tape
- Which is the speed of computer measured in?
 a) Nanoseconds b) Kilo-Seconds c) Gigahertz d) Megabytes
- Which of the following is not considered hardware?
 a) Operating System. b) CPU. c) Keyboard. d) Hard disk.

GROUP – 'B' [3×4=12]

Short Answer Questions.

- What do you mean by "Volatility"? Explain RAM and ROM with the concept and term. [1+3]
- What is a 'BUS' in computer architecture? Explain different types of 'BUS' in brief. [1+3]

'OR'

- Explain cash memory and Buffer Memory in details. [2+2]
- Differentiate between hibernate and sleep mode in PC. [4]

GROUP – 'C' [1×8=8]

Long Answer Questions.

- What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [2+6]

"Best of Luck"

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION- 2082

Class: XI
 Time: 1hr.

Morning Shift Set 'A'
 Subject: Computer Science [4271]

F. M.: 25
 P. M.: 10

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP – 'A' MCQ [5×1=5]

Multiple Choice Questions.

- The brain of Computer System is?
 a) ALU b) Memory c) Control Unit d) CPU
- Microprocessor was introduced in which generation of computer?
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- Which is the speed of computer measured in?
 a) Nanoseconds b) Kilo-Seconds c) Gigahertz d) Megabytes
- Which of the following is not considered hardware?
 a) Operating System. b) CPU. c) Keyboard. d) Hard disk.

GROUP – 'B' [3×4=12]

Short Answer Questions.

- What do you mean by "Volatility"? Explain RAM and ROM with the concept and term. [1+3]
- What is a 'BUS' in computer architecture? Explain different types of 'BUS' in brief. [1+3]

'OR'

- Explain cash memory and Buffer Memory in details. [2+2]
- Differentiate between hibernate and sleep mode in PC. [4]

GROUP – 'C' [1×8=8]

Long Answer Questions.

- What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [2+6]

"Best of Luck"

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2082

(Class: XI)
 Time: 2hrs.

MORNING SHIFT [SET-B]

Subject: English

F.M: 37

P.M: 15

Candidates are required to give their answer in their own words as far as practicable.
 The figure in the margin indicates full marks.

Attempt all questions.

1. Read the following text and do the activities that follow.

Long long ago, in the small village, people were terrified of an old abandoned mansion at the edge of the forest. Every night, strange noises echoed from inside—footsteps, whispers, and sometimes a chilling scream. Villagers first thought it was robbers or wild animals. But the weirdest thing was that the noises always happened exactly at midnight. Raj, a brave but lazy teenager, decided to investigate. Not because he wanted to explore, but because his younger sister dared him. He entered the mansion with only a flashlight and a small notebook.

As he stepped inside, the door creaked loudly behind him. Raj froze. Shadows flickered on the walls, and cold gusts of wind blew through the broken windows. Suddenly, he saw a pair of glowing eyes staring at him from the dark hallway. He quickly took a step back and realized it was a black cat—but then he heard a whisper saying, "Leave now..."

The next morning, a local historian explained that the mansion had been abandoned for decades after a family mysteriously disappeared. The noises were caused by the wind moving through broken floorboards and shutters, and the villagers' fear made the experience feel more terrifying than it really was.

From then on, people avoided the mansion at night, but some teenagers still dared each other to peek inside for a thrill.

A. Answer the following questions in a single sentence each: [5x1=5]

- What was strange about the abandoned mansion?
- Why did Raj decide to enter the mansion?
- How did the local historian explain the strange noises?
- What did Raj see first when he entered the mansion?
- What did people do after learning the truth about the mansion?

B. Fill in the blanks with suitable information from the passage. [5x1=5]

- People first thought the noises were caused by
- The noises always happened exactly at
- Raj entered the mansion with a flashlight and a small
- Raj saw a pair of glowing staring at him.
- The abandoned mansion was located at the edge of the

2. Explain with reference to the context: [5]

a) Stop crying, Oh man,
 Stop cursing me
 And listen.

*I didn't come here of my own free will.
 I was invited.*

*Believe me, I had no choice but to visit you.
 How many lives were lost
 Because of me? You count.
 But have you ever counted
 How many have died so far
 Because of you and your wars?*

3. Answer the following questions in about 75 words each. [3x2=6]

- Why did the Giant knock down the high wall around his garden? (*The Selfish Giant*)
- What is personification? Explain the use of it in the story, *The Selfish Giant*
- What human activities invited Corona to the world? (*Corona Says*)

4. Write a readable story using the clues given below: [8]

dog finds a piece of meat..... crosses a bridge..... sees his reflection.....
 thinks another dog has meat..... tries to snatch..... drops his own meat
 moral

5. Rewrite the sentences as indicated in the brackets. [3x1=3]

- He bought a car. (Put the adverb *recently* in the correct place.)
- She speaks English. (Put the adverb *fluently* in the correct place.)
- We play football. (Put the adverb *hardly* in the correct place.)

KATHMANDU MODEL SECONDARY SCHOOL
PRE-TERM EXAMINATION 2082

Class: XI

Time: 1.5 hrs.

Morning Shift: Set-A

Subject: Mathematics (0071)

F.M.: 37

P.M.: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
 Attempt all the questions.

GROUP-A [MCQ] [4×1=4]

Write the correct options of each question in your answer sheet.

1. Which of the following is the contrapositive of $p \Rightarrow q$?

- A) $p \Rightarrow \sim q$ B) $q \Rightarrow p$ C) $q \Rightarrow \sim p$ D) $p \Rightarrow q$

2. If $x \in (A \cup B)$, what does it mean?

- A) $x \in A$ and $x \in B$ B) $x \in A$ or $x \in B$
 C) $x \in A$ and $x \in B$ D) $x \in A$ and $x \in B$

3. What is the value of $\lim_{x \rightarrow 5} x$?

- A) e B) 5 C) undefined D) π

4. What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{x+4x^2}}{4+x}$?

- A) $\frac{1}{4}$ B) 4 C) 2 D) $\frac{1}{2}$

GROUP-B [5×5=25]

a) Write the negation of the statement:

"If $3 < 4$ implies that $3 > 7$ then 4 is odd".

Also, determine its truth value.

[1+1]

b) Construct truth table for the statement: $(p \wedge q) \wedge (\sim p \vee \sim q)$.

Also, what conclusion can be drawn from the truth table.

[2+1]

a) Define intersection of two sets.

[1]

b) List the various indeterminate forms that occur in limits.

[1]

c) Write the negation of $p \vee q$.

[1]

d) What is the value of $\lim_{x \rightarrow 0} \frac{x}{\sin x}$?

[1]

e) Define the complement of a set with an example.

[1]

7. State and prove De-Morgan's laws for two sets.

[1+4]

8. Evaluate: $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$

a) Using a table of values.

b) Using algebraic formula.

c) Is it possible to find $f(3)$? Explain.

[2+2+1=5]

9. a) If $A = \{a, b, c\}$ and $B = \{a, b\}$, verify that $P(B) \subset P(A)$.

[2]

b) Evaluate: $\lim_{x \rightarrow 9} \frac{x \sin x - 27 \sin y}{x^2 - 9}$

[3]

GROUP-C [8×1=8]

10. a) Does the order of elements in a set have any significance? Justify your answer with an example.

[2]

b) What is the difference between undefined and indeterminate forms in the context of limits? Justify your answer.

[3]

c) Give an example of a function which takes the indeterminate form $\frac{0}{0}$ but whose limit does not exist. Justify your answer.

[3]

Best of Luck

Class: XI

Time: 1hr.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP - 'A' [5×1=5]

Multiple Choice Questions.

1. 4 bits equals to:
a) 1 byte b) A nibble c) A block d) 2 nibbles
2. Which generation of computers introduced the use of integrated circuits (ICs)?
a) Third b) Second c) Fourth d) First
3. What is the primary function of cache memory in a computer system?
a) To run applications without needing an Internet connection
b) To store the operating system permanently
c) To provide long-term, archival storage
d) To act as a temporary holding area for frequently accessed data
4. MIPS stands for?
a) Multiple Instructions Per Second
b) Million Instructions Per Second
c) Major Instructions Per Second
d) Million Information Per Second
5. Which of the following is not considered output device?
a) Projector b) Screen c) Speaker d) microphone

GROUP - 'B' [3×4=12]

Short Answer Questions.

1. What do you mean by Primary memory? Differentiate Between SRAM and DRAM. [1×3]
2. What is CPU? Explain different components of CPU in brief. [1×3]

'OR'

Explain Address Bus and Data Bus in details.

[2+2]

3. Imagine you are designing a brochure for your school's annual fest. Using MS word, describe the steps of creating an eye catching cover page that includes:

[4]

- ❖ A catchy title
- ❖ An image related to the theme
- ❖ A slogan of the fest
- ❖ A background color or texture that complements the theme

GROUP - 'B' [3×4=12]

Long Answer Questions.

1. What do you mean by computer memory? Describe different types of computer memory with clear diagram. [2+6]

"Best of Luck"

1. ආගමික සංවිධාන සඳහා ප්‍රවේශයක් ලෙස ප්‍රවේශයක් ලෙස
 මානව හිමිකම් : මානව හිමිකම් මානව හිමිකම් මානව හිමිකම්
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1. 1998 年 2 月 21 日 1998 年 2 月 21 日 1998 年 2 月 21 日

[illegible]

1. "Zurück zum Leben" - ein Lied, das die Hoffnung auf ein besseres Leben ausdrückt.

1. 2000-2001 2002-2003 2004-2005 2006-2007 2008-2009 2010-2011

...the
... ..
... ..
... ..

1. The first part of the document is a list of names and addresses, which appears to be a directory or a list of subscribers. The names are written in a cursive script, and the addresses are listed below them.

(12) *Spine (spinal) 2 upper (upper) 2 lower (lower) 2*

1. The first thing I did was to go to the bank (1)

1. Mengingat bahwa kegiatan Zilaq Mahallada (M)

1. The applicant is not a citizen of the United States.

1. 2003-2004

1. Түркістан Республикасының Конституциясы мен Заңдарына сәйкес, Республикадағы барлық қандай да болсын мемлекеттік билік республикадағы барлық билік бірліктері мен билік бірліктерінің қолында болуы қажет.

[illegible]

($\supset = \text{A} + \text{A}$)

[illegible][illegible][illegible][illegible]

1. The principle of the work


KATHMANDU MODEL SECONDARY SCHOOL
 ८०८-११५४ फुल्लिहरे पर्व
 ०६ : फुल्लिहरे, न. न. न. पर्व
 १६ : फुल्लिहरे, न. न. न. पर्व

प्रश्नहरू :

- (क) केले गात्ता टुटाउँछ, केले गात्ता जोड्छ ?
 (अ) रिकले टुटाउँछ, पमले जोड्छ । (आ) खोरिले टुटाउँछ, माथले जोड्छ ।
 (इ) पुषाले टुटाउँछ, पैमले जोड्छ । (ई) लोभले टुटाउँछ, दामले जोड्छ ।
- (ख) हजारौं वर्ष पहिले कस्तो चलन थियो ?
 (अ) खोतीपाती लगन
 (इ) घर नबनाउने
 (आ) लुप्ता नलग्नुने
 (ई) एकजनाकोलाई सहयोग नगर्ने
- (ग) कोषी खानाको के भइसक्यो ?
 (अ) पोतेको खाना
 (इ) पकान्न
 (आ) उमानेको खाना
 (ई) भासी खाना
- (घ) रमका ठाउँमा के भइसक्यो ?
 (अ) रेल, मोटर, ट्याक्सी र हवाईजहाज
 (इ) रेल, मोटर, कार र मोटरसाइकल
 (आ) रेल, मोटर, हवाईजहाज र रकट
 (ई) रेल, मोटर, ट्याक्सी र रकट

५. प्रश्नको उत्तर लेख्नुहोस् ।

क) दिइएको कवितामा पढी सोधिएका प्रश्नहरूको दिनुहोस् :

(Y+Y=C)

दिग्गधोमा डोब तिपो थियो हिउँथि होला
 बेबाबती किनारभरि पौरखको थियो होला
 वीर पुर्खा । तिमीलाई भितेरिले माथै बोझो
 सागर तरी समारभरि वीर गोर्खा रगत बग्थो ।

प्रश्नहरू :

- (अ) माथिको कवितामा हाप्पा पुर्खाहरूका बारेमा के भनिएको छ ?
 (आ) हामीले हाप्पा पुर्खाको गौरव कमरी जोसाउन सक्छौं ।

घ) तलको कथा पढी सोधिएका प्रश्नको उत्तर दिनुहोस् :

खोला अबेल फेर गाउँतिर सोधिएको रहेछ । यो खोलाले पनि गाउँलाई दुम्कै रक्यो । हुन त रुवाथो माथै भन्नु खोलापति अन्याय ठहर्ने । यो खोलाले गाउँलाई पामेको पनि छ । हाप्पो बिगतको लगभग आधा उमेर यही खोलाको दिन र बगरहरूमा बितेको थियो । कहिले खोरहा खाने निर्हुमा त कहिले गाईपैनी चराउने निर्हुमा । खोलामा बाढी आएको बेला हामी गाउँको पुच्छर समाएर बाढी तर्ने गर्थौं अबका बस्ती बेलासा हामी खोलाको किनारमा बसेर बाढीको जानन्द मुटुप्यौं । तर्ने डराउने कोही मानिस आए भने हामी तीनचार जना एकजनाको हात समाउँदै बाढीमा होतप्यौं र किनारमा परिरेहको माँससलाई तारिदिन्यौं र बस्ती कसमा खान गरी परिछनकै विशेष रुचि थियो । "बर्हीनिर कुवा छ, गए मानको वर्षासा मूल कुटेको एकदम थियो र मिठो पानी छ ।" ऊ पानी लिएर आइसकेको थियो ।

प्रश्नहरू :

- (अ) खोलाले रक्यो भन्ने कुरा पात्र यही कसरी अन्याय हुन्छ ?
 (आ) "य" पात्रका खोलासँगका मानवपुष्टि कस्ता छन् ?

७. वीर पुर्खाहरूले कसरी वीरता देखाए ? सवीक्षायक उत्तर लेख्नुहोस् ।

C

सरलताको शुभकामना !!



Class: XI 'Science'
Time: 1.5 hrs.

MORNING SHIFT 'SET-B'
Subject: Physics [1011]

F.M: 37
P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.

GROUP - 'A' [6X1=6]

Multiple Choice Questions:

1. The equation of a wave is given by, $y = A \sin \left[\omega \left(\frac{x}{v} - k \right) \right]$, when ω is angular velocity x is position and v is linear velocity. The dimensional formula for k is:

- a) $[L T]$ b) $[T]$ c) $[T^{-1}]$ d) $[T^2]$

2. Identify the correct statement:

- a) Velocity, displacement, acceleration, displacement are vectors
b) Weight, mass, torque and energy are scalars
c) Angular velocity, momentum and electric current are scalars
d) A vector quantity is one which has magnitude only whereas scalar quantity is one which has both magnitude and direction.

When 2×10^{11} electrons are added to a neutral metal sphere, the charge on the sphere becomes:

- a) $-16 \mu C$ b) $16 \mu C$ c) $32 \mu C$ d) $-32 \mu C$

A soap bubble is given a negative charge, then its radius:

- a) Decreases b) Increases c) Remains unchanged
d) Nothing can be predicted as information is insufficient.

The force between two charges 0.06 m apart is 5 N. If each charge is moved towards the other by 0.01 m, then the force between them will be

- a) 7.20 N b) 11.25 N c) 22.50 N d) 45.0 N

When a positively charged, conductor is connected to earth:

- a) No charge flow occurs
b) Electrons flow from conductor to earth
c) Protons flow from conductor to earth
d) Electrons flow from earth to conductor

GROUP - 'B' [5X3=15]

Short Answer Questions:

- a) What are the basic rules of dimensional analysis?

[1]

- b) Can we tell the unit of a physical quantity from its dimensions? What about dimensions from the unit of a physical quantity? [1]

- c) Check the correctness of the physical equations: (i) $t = 2\pi \sqrt{\frac{l}{g}}$; t is time period, l is length and g is acceleration due to gravity (ii) $T^2 = \frac{4\pi^2 a^3}{GM}$; T is time period, G is gravitational constant, M is mass and a is radius of the orbit. [3]

8. a) What is meant by resultant vector. [1]
b) Two vectors acting on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangular representation? State and explain the law involved. [3]
c) Under what condition is the magnitude of resultant vector is zero vector? [1]
9. a) Explain about Action of point with a demonstration? [2]
b) What do you mean by electrostatic induction? How will you give permanent positive charge to a conductor by the method of induction? Explain in detail. [3]

GROUP - 'C' [2X8=16]

Long Answer Questions

10. a) What do you mean by dimension of a physical quantity? What is the difference between dimensions and unit of a physical quantity? [2]
b) A body moving through air at a speed v experiences a retarding force F given by the equation $F = k A \rho v^x$, where A is the surface area of the body, ρ is the density of air and k is dimensionless constant. Deduce the value of x . [2]
c) Convert 60 J/s into a new system of unit which has 10 gm, 10 cm and 10 second as fundamental units of mass, length and time. [2]
d) Is it possible to derive all the physical equations by the method of dimensional method? Explain. [2]
11. a) Write the formula for Coulomb's force in vector form? What is the significance while expressing in vector form? [2]
b) What is electrical permittivity? Write its value and unit for vacuum in SI and cgs system of unit. [2]
c) If a body gives out 10^9 electrons per second, how much time is required to get a total charge of 1 C from it? What conclusion would you draw from the result obtained? [2]
d) Two positive point charges are at a separation of 0.3 m repel each other with a force of 1.6 N. If the sum of two charges be 10 microcoulomb, what are their separate values? [2]

Best of Luck

Time: 1:30 hrs.
 Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

MULTIPLE CHOICE QUESTIONS

GROUP - 'A' [9×1=9]

1. When zinc is added to CuSO_4 solution, copper is precipitated. It is so because,

- a) Reduction of Zn
 b) Reduction of Cu^{+2}
 c) Hydrolysis of CuSO_4
 d) Reduction of SO_4^{2-}

Which of the following acts as oxidizing agent only?

- a) NO_2
 b) N_2O_5
 c) HNO_3
 d) NH_3

Oxidation number of Hydrogen in MgH_2 is

- a) +1
 b) -1
 c) +2
 d) -2

The increase in volume that will take place when 200 ml of gas at 300 K is heated to 360 K at constant pressure is

- a) 240 ml
 b) 200 ml
 c) 40 ml
 d) 160 ml

A gas which perfectly obeys gas laws at all temperature is called

- a) Noble gas
 b) Ideal gas
 c) Inert gas
 d) diatomic gas

Value of 'R' in Joule is

- a) 8.314
 b) 1.987
 c) 0.0821
 d) 8.314×10^7

Which of the following transition involves maximum amount of energy?

- a) $M^-(g) \rightarrow M^+(g)$
 b) $M(g) \rightarrow M^+(g)$
 c) $M^+(g) \rightarrow M(g)$
 d) $M(g)^{+2} \rightarrow M(g)^{+3}$

The correct order of ionization energy of C, N, O, F is

- a) $F > N > O > C$
 b) $O > N > F > C$
 c) $O > N > C > F$

d) $F > O > N > C$

SHORT ANSWER QUESTIONS

GROUP - 'B' [5×4=20]

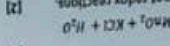
9. Isoelectronic structure from the following is

- i) CH_3^+
 ii) H_2O^+
 iii) NH_4^+
 iv) CN^-
 a) (i) & (ii)
 b) (iii) & (iv)
 c) (i) & (iii)
 d) (ii) & (iv)

LONG ANSWER QUESTION

GROUP - 'C' [1×8=8]

1. a) Define electrocration and de-electrocration with examples. [2]
 b) Find the maximum and minimum oxidation number of Bromine and Phosphorous with example. [2]
 c) Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent. [2]
 d) Show with examples that all reactions are not redox reactions. [2]



Best of Luck



KATHMANDU MODEL SECONDARY SCHOOL

PRE-TERM EXAMINATION 2082

Class: XI

Time: 1.5 hrs.

Morning Shift Set-B
Subject: Mathematics (0071)

P.M.: 37

P.M.: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [4×1=4]

Rewrite the correct options of each question in your answer sheet.

- What is the contrapositive of $q \Rightarrow \sim p$?
A) $p \Rightarrow \sim q$ B) $q \Rightarrow p$ C) $q \Rightarrow \sim p$ D) $p \Rightarrow q$
- Which one of the following is true?
A) $A \subseteq A \cup B$ B) $A \cap B \subseteq B$ C) $A \cup \bar{A} = U$ D) All of the above
- What is the value of $\lim_{x \rightarrow 5} -1$?
A) 1 B) -1 C) undefined D) 5
- What is the value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+25x^2}}{3+5x}$?
A) $\frac{1}{5}$ B) 5 C) 2 D) 1

GROUP-B [5×5=25]

- Determine the truth value of the statement "3 is odd and 4 is even". Also, write its negation. [2]
 - Construct truth table for the following statement and comment on the result $(p \Rightarrow \sim q) \wedge (p \wedge q)$ [3]
- Is it possible to write $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{1}{x} - \lim_{x \rightarrow 0} \frac{1}{x}$? Give reason. [2]
 - Evaluate:
 - $\lim_{x \rightarrow 4} \frac{\sec y - \sec x}{\sqrt{x} - \sqrt{y}}$
 - $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$ [2+1]
- Define limit of a function $f(x)$ at $x = c$. [1]
 - What are indeterminate forms? [1]
 - Define union of two sets. [1]
 - State De-Morgan's laws for logic. [1]
 - What is the value of $\lim_{y \rightarrow 0} \frac{y}{\tan y}$? [1]

- For any two non-empty subsets P and Q of universal set U , prove that

$$\overline{P \cap Q} = \overline{P} \cup \overline{Q}$$
 [2]
- If $A \subseteq B$, then $P(A) \subseteq P(B)$. Verify this by choosing appropriate sets A and B . [3]
- For any three sets A, B , and C , prove that:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
- Evaluate: $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{\sqrt{4x}-\sqrt{12+4}}$ [2+3]

GROUP-C [8×1=8]

- Does the order of elements in a set matter? Justify your answer with an appropriate example. [2]
 - What is the difference between undefined and indeterminate forms in the context of limits? Justify your answer. [3]
 - Give an example of a function that takes the indeterminate form $\frac{0}{0}$ but whose limit exists. Justify your answer. [3]

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL

पूर्व त्रैमासिक परीक्षा-२०८२

कक्षा : ११

समय : १:३० मिनेट

विभागीय सत्र 'क'

विषय : नेपाली (००११)

पूर्णांक : ३०

उत्तीर्णांक : १५

विद्यार्थीहरूले तल्लेखमा आफ्नै हातमा उत्तर दिनुपर्ने छ । यहाँ कित्तामा दिइएको अङ्कले पूर्णांक ज्ञात गरिनेछ ।

यसै प्रश्नहरू अनिवार्य छन् ।

१. दिइएका व्यञ्जन वर्णहरूको उच्चारण स्थान र प्रागल्भ छुट्याई लेख्नुहोस् :

य, म, ह

२. दिइएको वाक्यलाई शुद्ध गरी पुनर्लेखन गर्नुहोस् :

बाल्यकालको दसैँको स्मृति मेरो हृदयमा अर्थात् सम्म तान्ने छ । म पूर्वी नेपालको पहाडमा जन्मेको छिर् । मेरो गाउँ भन्दा अर्थात् दसैँको यकल सम्झुला जम्मो र त्यसै रमाउँछो ।

३. दिइएको अनुच्छेदमा रेखाङ्कित पदहरूको पदवर्ग पहिचान गरी लेख्नुहोस् :

उसको नजानु हेराइ, लचकदार बिँदाइ, मिर्जासिलो स्वभाव सबै उत्तिकै राम्रो लाग्छ त्यसैले ऊ सरासर घर गएका बेला पनि मैले उसको बाटो हेरेको छु ।

४. दिइएका वाक्यहरूलाई बहुवचनमा परिवर्तन गरी पुनर्लेखन गर्नुहोस् :

म कक्षा १२ मा पढ्ने विद्यार्थी हुँ । कलेजमा मेरो एकजना मन भिन्ने साथी छ । साथी मलाई माया माथ होइन सहयोग पनि गर्छ । म दिनहुँ उसलाई भेट्छु ।

५. दुवै प्रश्नहरूको उत्तर दिनुहोस् :

(Y+Y= ८)

क. दिइएका अनुच्छेद पढी सोधिएका प्रश्नहरूको उत्तर दिनुहोस् :

आयुर्वेद पूर्वीय प्राचीन चिकित्सा पद्धति हो । 'आयुर्वेद' शब्दको अर्थ 'आयुको ज्ञान' भन्ने हुन्छ । यसलाई आयुर्विज्ञान पनि भनिन्छ । आयुको ज्ञान भन्नाले जीवनमा पुर्ण आयु जित्ने विषय र कलाको ज्ञान हो । पुर्ण आयु जित्न स्वास्थ्य पक्षको आवश्यकता हो । स्वस्थ व्यक्ति नै पुर्णायुको जीवन जिउँछ भन्ने आयुर्वेदको मान्यता हो । यसैले आयुर्वेदमा जीवन स्वस्थ राख्ने उपायका बारेमा विस्तृत रूपमा चर्चा गरिएको छ । जीवन स्वस्थ राख्नु भनेको सक्तर भर शरीरमा रोगको प्रवेश हुन नदिनु हो र कदाचित् रोगको प्रवेश भइहाले पनि उपचारका माध्यमबाट शरीरबाट त्यसको निष्कासन गर्नु हो । आयुर्वेदमा रोगको उपचारका लागि कार्वाचिकित्सा, शान्त तन्त्र, शास्तास्य तन्त्र, भूतविद्या, कौमारभृत्य, अगद तन्त्र, रसायन तन्त्र र बाजीकरण तन्त्र गरी आठ अङ्ग रहेका छन् । बीचमा कार्वाचिकित्साको शारीरिक रोग, शान्त तन्त्रमा चिरकर गर्नुपर्ने रोग, शास्तास्य तन्त्रमा नाक, कान, घाँटी वा अन्य कुनै कारण भएको टाउकाको रोग, भूतविद्यामा भूत, गृह वा मानसिक कमजोरीका कारण लागेको मानसिक रोगको उपचारको पद्धति प्रस्तुत गरिएको छ । यसैले कौमारभृत्यमा गर्भवती स्त्रीलाई तान्ने रोग र बालरोग, अगद तन्त्रमा

शरीरमा विषाणु पदार्थको प्रवेशका कारण उत्पन्न भएको रोग, रसायन तन्त्रमा कुष्ठरोगमा हुने रोग, बाजीकरण तन्त्रमा शरीरमा घातको कटौका कारण हुने रोगको उपचारको पद्धति समर्पित छ । आयुर्वेदका अनुसार दुस्रो रूपमा शरीरमा वात, पित्त र कफ यी त्रिदोषको असन्तुलनका कारण रोगको प्रवेश हुन्छ । यी तीन दोषबाट बच्ने मुख्य उपाय भनेको उचित आहार र उपयुक्त जीवनशैली नै हो । उपयुक्त आहार र जीवनशैलीबाट शरीरमा रोगसँग अड्कने खोज वा जीवनी शक्ति बसाल्न रूपमा चिकित्सीय बन्छ । जीवनी शक्तिको तत्त्वज्ञान हुँदा शरीरमा रोगले प्रवेश नै गर्न पाउँदैन, कदाचित् प्रवेश गरिहाले पनि त्यो प्रभावहीन हुन्छ । प्रभावहीन तन्त्र रोगलाई पनि स्वाधीन जडीबुटी र सन्तानको उपयोग गरी शरीरबाट सड्ने निष्कासन गर्ने गरिन्छ । यसैले उचित आहार, उपयुक्त जीवनशैली र उपलब्ध स्वाधीन जडीबुटीको सेवनबाट स्वस्थ रहेर पुर्णायुको जीवन जित्न सकिन्छ भन्ने कुराको मार्गनिर्देशन गर्ने चिकित्सा पद्धति नै आयुर्वेद हो ।

प्रश्नहरू :

- आयुर्वेदका आठ अङ्ग के के हुन् ?
- कार्वाचिकित्सा र शान्त तन्त्रमा कुन कुन रोगको उपचार पद्धति प्रस्तुत गरिएको छ ?
- शरीरमा रोगको प्रवेश हुनुको कारण के हो ?
- आयुर्वेदले जीवनमा पुर्णायु जित्न के गर्नुपर्ने सुझाव दिएको छ ?

ख. दिइएको कवितामा पढी सोधिएका वस्तुगत प्रश्नहरूको उत्तर लेख्नुहोस् :

आँसु छताली रोइरहेपे जीवन वनका बिच
छोटी छोटी कतै नपाई मेरो मनका बिच
टुटेभुटेका आशानाथि नवन दुखका गौर
कहाँ कहाँको कला कलाको मेरो मनको गिर ।
नालमाते मोहनको रूप गरी धारण बोयी
गहना भन्ज्याई रुम्छी तिनी कुन कारण ।
चित्तमाको यी महल मनोहर हुन्छ
म तिमी गानी मैले बोले,
"हुन्छ सबै यो भोति भुत्ने र खानी ।"
धर्म बोले, "कुन मन्दिरमा पुजा गर्छौ आरु"
मैले बोले, "शरीरहरूको बलिदान भएको ठाउँ"
सहानुभूतिले सङ्करण बोलेन, "तब किन रुम्छौ भाइ ?"
साथ छ को को ? घर हो तिमी कुन देश ?
जान्छौ कुन पुर भवन मुसाफिर ? ज्यामी कुन सन्देश ?
"रोल्छ छ मेरो मुप उचोरी, साव छ साहस बेस
विश्व सबै घर हृदयपुरीतिर ज्याउँछु सेवा सन्देश ।"
प्रश्नहरू :

-सन्धीपमा देवकोटा

Class: XI 'Science'
 Time: 1:30 hrs.
 Date: 15/01/2018
 Subject: Chemistry [3011]
 P.M: 15
 F.M: 37

Attempt all the questions.

MULTIPLE CHOICE QUESTIONS
 GROUP - 'A' [9x1=9]

1. A reducing agent is a substance which can
 - a) accept electron & is oxidized
 - b) donate electron & is oxidized
 - c) accept electron & is reduced
 - d) donate electron & is reduced
- Which of the following acts as reducing agent only?
 - a) SO_2
 - b) H_2SO_4
 - c) H_2S
 - d) SO_3
- 16 gm of O_2 at NTP would occupy a volume of
 - a) 11.2 lit
 - b) 22.4 lit
 - c) 5.6 lit
 - d) 16 lit
- The density of NH_3 at $30^\circ C$ and 5 atm pressure is
 - a) 2.30 gm/lit
 - b) 3.42 gm/lit
 - c) 23 gm/lit
 - d) 30 gm/lit
- The curve plotted between P and V at constant temperature is called
 - a) Isotherm
 - b) Isobar
 - c) Isochore
 - d) Isomer
- The size of following species increase in the order
 - a) $Mg^{+2} < Na^+ < F^- < Al$
 - b) $F^- < Al < Na^+ < Mg^{+2}$
 - c) $Al < Mg^{+2} < F^- < Na^+$
 - d) $Na^+ < Al < F^- < Mg^{+2}$
- The outmost electronic configuration of the least electronegative element is
 - a) $ns^2 np^5$
 - b) $ns^2 np^4$
 - c) $ns^2 np^3$
 - d) ns^1
- The correct order of ionization energy of C, N, O, F is
 - a) $F > O > N > C$
 - b) $O > F > N > C$

SHORT ANSWER QUESTIONS
 GROUP - 'B' [4x5=20]

1. a) Define the terms:
 - i) Effective Nuclear Charge
 - ii) Periodicity in Periodic Table
- b) Define ionisation potential. Describe any two factors that affect ionisation potential.
- a) Define isoelectronic species.
- b) Compare size of (i) Na with Na^+ (ii) Cl with Cl^-
- c) Why is the electron affinity of chlorine higher than that of fluorine?
- a) Derive combined gas equation.
- b) Find the value of R' in SI unit.
- c) What is the significance of R'?
- a) Write any two applications of Boyle's law.
- b) A spherical balloon of 21 cm diameter is to be filled with hydrogen gas at NTP from a cylinder containing gas at 20 atm and $27^\circ C$. If the cylinder can hold 2.82 litre of water vapour at NTP, calculate the number of balloons that can be filled up.

LONG ANSWER QUESTION
 GROUP - 'C' [1x8=8]

1. a) Differentiate between oxidation number and valency.
- b) Define auto-redox reaction with example.
- c) Identify oxidation and reduction reaction as well as oxidizing agent and reducing agent.
- d) Find the oxidation number of
 - i) S in H_2SO_4
 - ii) Cr in CrO_3

"Best of Luck"

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PRE-TERM EXAMINATION- 2082
Day Shift Set 'A'

Class: XI
Time: 1hr.

Subject: Computer Science [4271]
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

F. M.: 25

P. M.: 10

GROUP - 'A' [5×1=5]

Multiple Choice Questions.

What is smallest unit of the information?

- a) A bit b) A byte c) A block d) A nibble

Which generation of computers saw the development of minicomputers and the emergence of time-sharing operating systems?

- a) First b) Second c) Fourth d) Third

Which of the following is typically used to store the computer's boot-up instructions and firmware?

- a) Read-Only Memory (ROM) b) Random Access Memory (RAM)
c) Hard Disk Drive (HDD) d) Solid State Drive (SSD)

A CPU with a clock speed of 3.5 GHz can perform how many clock cycles per second?

- a) 3.5 million cycles b) 3.5 billion cycles
c) 3.5 trillion cycles d) 350 million cycles

A device that receives data from the computer and displays it in a human-readable form is called a(n) _____ device?

- a) Processing b) Storage c) Input d) Output

GROUP - 'B' [3×4=12]

Short Answer Questions

What is ROM? Explain different types of ROM in details.

[1+3]

How do you define term Hardware, Software and Firmware? Give one example of each.

[3+1]

'OR'

Differentiate between Analog and Digital Computer.

[4]

3. Write down the steps to create folder and file in Windows Operating system.

[4]

GROUP - 'C' [1×8=8]

Long Answer Questions.

1. What do you mean by generation of computer? Describe different generations with all the main features included in each.

[2+6]

"Best of Luck"

- d) What warnings does corona give to human beings? Explain. (Corona Says)
- Write a readable story using the clues given below:
- cats – find a piece of bread ... quarrel over it ... monkey offers to divide – eats a little from each side to make equal ... finishes bread – cats fooled – moral
- 3 Rewrite the sentences as indicated in the brackets.
- 3a) I have bought the car. (Put the adverb *already* in the correct place.)
- 3b) She looks pretty. (Put the adverb *fairly* in the correct place.)
- 3c) They visit America. (Put the adverb *now and again* in the correct place.)

[3x1=3]

Choose the best options:

- a) The prefix 'un-' goes with the word ...
 i) happy ii) legal iii) possible iv) correct
- b) The meaning of the prefix in the word 'misuse' is
 i) before ii) again iii) wrong iv) not
- c) The dress is very **elegant**. Choose the correct meaning of the word 'elegant'.
 i) clumsy ii) beautiful iii) heavy iv) dirty
- d) If the guide words are "**crop – crush**", which of the following will appear on that page?
 i) cry ii) cross iii) crust iv) crow
- e) 'Stone' comes before 'Story' because comes before in the alphabet. (Fill in the blanks with suitable letters)
 i) n/r ii) r/n iii) e/y iv) y/e

[Best Wishes]

Students are required to give their answers in their own words as far as possible. The figures in margins indicate full marks.
 Page: XI
 Time: 1hr.
 P. M.: 10
 F. M.: 25

GROUP - 'A' [5×1=5]

Multiple Choice Questions.

- How many nibble(s) equals to 1 byte?
 a) 4
 b) 1
 c) 2
 d) 1024
- The introduction of the graphical user interface (GUI) and the mouse is a key feature of which generation?
 a) Fifth
 b) Second
 c) Fourth
 d) Third
- Short Cut key to rename the file's name is:
 a) F1
 b) F2
 c) F3
 d) none of these
- Why is the 'word size' (e.g., 32-bit vs. 64-bit) a factor in a computer's speed?
 a) It determines the physical size of the CPU chip
 b) It dictates the type of monitor that can be connected
 c) It controls the speed of the internet connection
 d) A larger word size allows the CPU to process more data in a single operation
- Which of these devices can be used for both input and output?
 a) Webcam
 b) Scanner
 c) Plotter
 d) Touchscreen

GROUP - 'B' [3×4=12]

Long Answer Questions.

- [1+3] What is Computer Bus system? Explain its types.
 [2+2] What is Explain the difference between a margin and a page break. Where would you find the options to adjust these in Word?
 [4] Explain the features and benefits of using MS word for document creating and editing.

Long Answer Questions.

GROUP - 'C' [1×8=8]

3. What is Mobile Computing? Explain its advantages and usability. [2+2]
 1. Explain the evolution of computer, describing the technologies used in different generations of computer.
 [8]

"Best of Luck"

Answers are required to give the answer in their own words as far as practicable.
 Figure in the margin indicates full marks:
 Time: 2hrs.

Attempt all questions.

Read the following excerpt and do the activities that follow.

Alyohin often thought of Anna Alexeyevna, the young, intelligent, and graceful wife of Luganovitch. They had met at a dinner in the town, shortly after a legal case had exhausted Alyohin. She was striking, with cordial, intelligent eyes that made him feel immediately close and familiar. Her charm and warmth impressed him deeply, and though she was married, a gentle connection began to grow between them.

During the summer, Alyohin spent long hours at Sofino, the estate where he worked. Though busy with farming and managing the estate, his thoughts often drifted to Anna Alexeyevna. When he visited the town, he always went to see her. She would greet him anxiously, asking why he had been absent, and her eyes, her voice, her step, always made him feel something extraordinary. They talked for hours, sometimes in silence, sometimes at the piano, sometimes playing with her child, yet never confessing the love that grew between them.

Alyohin struggled with his feelings, thinking of her husband, her children, and the life they shared. He knew that to express his love would disrupt their household and bring dishonor. Anna Alexeyevna seemed to reason the same way. They both silently reassured the moments together, yet their affection remained hidden, guarded by propriety and caution.

Years passed. Anna Alexeyevna became more withdrawn, suffering from low spirits and neurasthenia, while Alyohin continued to visit. They sat side by side at the theatre, shoulders touching, feeling the closeness of hearts unspoken. At last, the day of parting came. At the train station, Alyohin ran to her compartment to return a forgotten basket. Their eyes met, and all restraint vanished. She pressed her face to his chest, tears flowed freely, and he confessed his love. The train moved on. He kissed

A. Answer the following questions in a single sentence each: [5X1=5]

a) Who was Anna Alexeyevna?
 b) How did Alyohin feel when he first saw Anna Alexeyevna?
 c) Why did not Alyohin and Anna Alexeyevna openly confess their love?
 d) How did Alyohin and Anna Alexeyevna spend time together?
 e) How did Alyohin express his feeling for Anna at the train station?

B. Fill in the blanks with suitable information from the passage: [5X1=5]

a) Alyohin first met Anna Alexeyevna at a _____ hosted by her husband.
 b) Alyohin often spent long hours at _____ thinking of Anna Alexeyevna.
 c) At the theatre, they used to sit side by side, with their _____ touching.
 d) Anna suffered from low spirits and was being treated for _____ in her later years.
 e) At the train station, Anna had almost forgotten a _____ [5]

2. Explain with reference to the context: [5]

a) You call yourself 'the crown of creation';
 What about the others.
 Those who fly in the sky, Those who live in the ocean,
 And those, the sources of your oxygen?
 You think they all are your slaves
 Who you can sell or kill
 At your will.

3. Answer the following questions in about 75 words each. [3X2=6]

a) What are symbols? Describe the symbols used in the story, *The Selfish Giant*.
 b) Describe the Giant's garden. (*The Selfish Giant*)

Students are required to give their answer in their own words as far as practicable.
No figure in the margin indicates full marks.

Attempt all questions.

Read the following text and do the activities that follow.

It is impossible to say how first the idea entered my head, but once it did, I knew I could not rest until it was carried out. You see, there was an old man in the house where I lived. He was kind and gentle, and had never wronged me in any way. Yet his eye—oh, his eye!—it was like the eye of a vulture, pale blue, with a film over it. Whenever he looked at me, my blood ran cold. I decided that I must rid myself of that eye forever. Every night, I crept into his room as he slept, careful not to awaken him. I moved silently, like a shadow, listening to his even breathing. For seven nights, I could not kill him as his eyes were closed. At first, the task seemed impossible, but I waited patiently. Finally, on the eighth night, I saw my chance. The old man's eye was open, staring at me with that ghastly glare. My heart pounded wildly, yet I moved with the utmost care. Suddenly, a faint sound made me pause—it was the sound of his heartbeat, growing louder and louder in my ears. The fear, the suspense, was unbearable. In a frenzy of horror and obsession, I killed him. I struck swiftly, silencing the eye forever. The old man lay still. At first, I felt relief, then a strange unease took hold. I hid the body carefully under the floorboards, confident that no one would ever discover my deed. But soon, the sound began again. A dull, steady thumping filled my mind. At first, I told myself it was my imagination, but the sound grew louder, echoing in my skull. It was the beating of the old man's heart! No one could hear it, yet it tortured me. In the end, I could bear it no longer. Driven mad by guilt, I confessed everything to the authorities.

- A. Answer the following questions in a single sentence each.** [5x1=5]
- Why did the narrator decide to kill the old man?

- Why was the narrator unsuccessful in killing the old man for seven nights?
- Where did the narrator hide the old man's body?
- Why did the narrator confess to the authorities?
- Write a suitable title for the passage.

B. Fill in the blanks with suitable information from the passage. [5x1=5]

- The old man had never _____ the narrator in any way.
- The eye of the old man was pale blue, like the eye of a _____.
- The narrator crept into the old man's room every night while he was _____.
- The narrator moved like a _____ when he entered the old man's room.
- The narrator heard a _____ sound that filled his mind.

2. Explain with reference to the context:

- I will depart one day.
But remember
There're many others like me.
They'll come too.
If you don't get rid of your inflated ego,
You'll be back to your cave time
That you endured
Long, long, long ago —

3. Answer the following questions in about 75 words each. [3x2=6]

- What is the theme of the story, *The Selfish Giant*? Describe.
- Who is the little child compared to? Discuss. (*The Selfish Giant*)
- What message does Corona give to humankind? (*Corona Says*)

4. Write a readable story using the clues given below:

- two friends walking through forest — bear comes — one climbs tree — other lies down — bear sniffs and goes away — friend on tree asks what bear whispered — "Never trust a selfish friend." — moral

[8]

Rewrite the sentences as indicated in the brackets.

4. She has finished her homework. (Put the adverb *just* in the correct place.) [3x1=3]

5. The boy is intelligent. (Put the adverb *fairly* in the correct place.)
6. We go for a walk. (Put the adverb *rarely* in the correct place.)

Choose the best options:

7. The prefix 'im-' goes with the word —
i) patient ii) legal iii) correct iv) happy
8. The meaning of the prefix in the word '**repaid**' is —
i) wrong ii) again iii) before iv) not
9. The food was very **delicious**. Choose the correct meaning of the word '**delicious**'.
i) tasteless ii) mouthwatering iii) dirty iv) spicy
10. If the guide words are '**moon – more**', which of the following will appear on that page?
i) mouse ii) morning iii) move iv) moral
11. 'Plant' comes before 'Plate' because — comes before — in the alphabet. (Fill in the blanks with suitable letters)
i) n/t ii) e/n iii) a/e iv) e/t

[Best Wishes]

Students are required to give their answers in their own words as far as practicable.
 Figures in the margin indicate full marks.
 Report all the questions.

GROUP - A [MCQ] (4x1=4)

1. Which one of the following is logically equivalent to $\sim (p \vee \sim q)$?
 A) $p \wedge q$ B) $\sim p \wedge \sim q$ C) $\sim p \vee \sim q$ D) $p \wedge q$
 2. What is the value of $\lim_{x \rightarrow \infty} \frac{x^2 + 16x + 4}{x}$?
 A) 4 B) 2 C) 4 D) 8

3. Which one of the following is true?
 A) $A \cap B \subseteq A$ B) $A \cap B \subseteq A$ C) $A \cap A = \phi$ D) all of the above
 4. What is the value of $\lim_{x \rightarrow y} \frac{x^2 - y^2}{x - y}$?
 A) undefined B) 0 C) undefined D) y

Define intersections of two sets with an example.

Define contradiction of a statement.

What is the value of $\lim_{x \rightarrow 0} \frac{x^2 \ln(1+y)}{x}$?

What the value of $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x}$?

Write the negation of $p \Rightarrow q$.

[1+1+1+1+1=5]

12. a) $\lim_{x \rightarrow 0} \frac{x^2 \tan(x+y) - \sqrt{\tan x}}{x}$

b) $\lim_{y \rightarrow 0} \frac{\frac{x-y}{\sqrt{x-y}}}{\frac{x}{\sqrt{x-y}} - \sqrt{x-y}}$

[2+3]

Define set. How does it differ from collection? Justify your answer. [2]
 For any two non-empty subsets M and N of universal set U, prove

i) $M \cup N = M \cup N$
 ii) $M \cap N = M \cap N$

[1.5+1.5=3]

GROUP - C (8x1=8)

11. a) Is it possible to write $\lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} [f(x) \times g(x)]$? Explain with an example.

b) Under what condition are two statements logically equivalent? Justify your answer. [3]
 c) Does the order of objects in a set matter? Explain. [2]

comment on the result.

10. a) Construct truth table for the statement $(p \Rightarrow q) \vee (p \wedge \sim q)$ and

b) Define power set with an example.

c) Evaluate $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x} - \sqrt{x-4})$

d) but whose limit does not exist.

9. a) Give an example of a function which takes the indeterminate form $\frac{\infty}{\infty}$.

b) Is it possible to find $f(3)$? Explain.

c) Using algebraic formula.

d) Using a table of values.

8. Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 4}{x - 2}$

[2+2+1+5]

Best of Luck

Advancing Education Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
 PRE-TERM EXAMINATION 2082

Page: XI

Time: 1.5 hrs.

Day Shift SET-A

Subject: Mathematics (0071)

F.M.: 37

P.M.: 15

Students are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
 Attempt all the questions.

GROUP - A [MCQ] [4×1=4]

Write the correct options of each question in your answer sheet.

Which of the following is logically equivalent to $\sim (p \wedge \sim q)$?

- A) $\sim p \wedge q$ B) $\sim p \vee q$ C) $\sim p \vee \sim q$ D) $p \vee \sim q$

If $A = \{a, b, c, d\}$, then the total number of nonempty proper subsets is

- A) 15 B) 14 C) 16 D) 13

What is the value of $\lim_{x \rightarrow 0} \frac{x^2}{x^3}$?

- A) 0 B) $\frac{1}{0}$ C) ∞ D) $-\infty$

What is the value of $\lim_{x \rightarrow 2} \sqrt{2} = ?$

- A) $\sqrt{2}$ B) 2 C) undefined D) $1/\sqrt{2}$

GROUP - B [5×5=25]

a) Define difference of two sets.

b) State De- Morgan laws for logic.

c) What is the value of $\lim_{x \rightarrow 0} \frac{\tan(x-\theta)}{x+\theta}$?

d) What the value of $\lim_{x \rightarrow 0} \frac{2^x - 1}{x}$?

e) Define conditional statement.

[1+1+1+1+1=5]

evaluate: a) $\lim_{x \rightarrow 0} \frac{\sqrt{2x+1} - \sqrt{x+1}}{\sqrt{x}}$

b) $\lim_{y \rightarrow 0} \frac{\sqrt{\sin(x+y)} - \sqrt{\sin x}}{y}$

[2+3]

f) Define set. How does it differ from collection? Justify your answer. [2]

g) For any two non-empty subsets X and Y of universal set U , prove that:

i) $\overline{X \cap Y} = \overline{X} \cup \overline{Y}$

ii) $\overline{X \cup Y} = \overline{X} \cap \overline{Y}$

[1.5+1.5=3]

10. a) If $f(x) = \frac{\sin(x-a)}{x^2-a^2}$, then

i) Determine $f(a)$, if it exists. [1]

ii) Examine whether $\lim_{x \rightarrow a} f(x)$ exist? Justify your answer. [1]

b) Evaluate: $\lim_{x \rightarrow \sqrt{5}} \frac{x^2-5}{x^2+5\sqrt{5}x-30}$ [3]

11. a) Define power set with an example. [2]

b) Construct truth table for the statement $(p \vee q) \vee (\sim p)$ and comment on the result. [3]

GROUP - C [8×1=8]

12. a) Is it possible to write

$$\lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} [f(x) + g(x)]?$$

Explain with an example. [3]

b) Under what condition are two statements logically equivalent? Justify your answer. [3]

c) Does the repetition of elements in a set affect the set? Explain. [2]

Best of Luck

[illegible]

॥ श्रीगणेशाय नमः ॥

1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840. 84

: 222222

1. These are the best of the city and the state

ਭਾਗ ੬ : ਪੰਨਾ ੧੭੭

பெரிய கிணறு

1872 1873 1874 1875 1876 1877

3) മലയാളം ക്രമരേഖകൾ ക്രമവ്യവസ്ഥാപനം സൂചനാപത്രം ക്രമരേഖകൾ (3)

($\Sigma = \lambda + \lambda$)

॥ श्रीगणेशाय नमः ॥

21 (5)

12k (2)

Exh. (11c)

124.45 (s)

இந்தக் கட்டுரை எழுதின அமைச்சர் (B)

212-4E (2)

上島 (島)

Ball, (16)

INDEX (b)

2. ଜଣେ ଶ୍ରମିକଙ୍କୁ ଦିଆଯାଇଥିବା ମାସିକ ମାଗଣା

h1h2 (2)

11-1215 (2)

1991-1992

1981-1982 (1981)

6. The following are the names of the members of the committee:

Infile (5)

附錄二 (續)

1990

1992

2. How the risk of the business is shared

1144

История вопроса

[illegible]

(122)

(Note: The text is extremely faint and appears to be a list of names or a table of contents. The visible text includes "The first of the..." and "The second of the...".)

2. 2015. 12. 15.

[illegible][illegible]

[illegible]

1. The first part of the document is a list of names and addresses, which appears to be a directory or a list of contacts. The names are written in a cursive script, and the addresses are listed below them. The list includes names such as "John A. Smith", "John B. Smith", "John C. Smith", "John D. Smith", "John E. Smith", "John F. Smith", "John G. Smith", "John H. Smith", "John I. Smith", "John J. Smith", "John K. Smith", "John L. Smith", "John M. Smith", "John N. Smith", "John O. Smith", "John P. Smith", "John Q. Smith", "John R. Smith", "John S. Smith", "John T. Smith", "John U. Smith", "John V. Smith", "John W. Smith", "John X. Smith", "John Y. Smith", and "John Z. Smith".

(1) *Indefinite pronouns* are used to refer to people or things in general.

(1) The following information is required to be submitted to the Commission:

1. *Therapsid* (mammals) (In)

1. The first step is to identify the problem or question that needs to be answered.

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Kopie des Originals

... ..

1. The first step is to identify the problem or question that needs to be answered. This involves understanding the context and the specific information required.

1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 2680, 26

[illegible]
$$(\sup = \lambda + \lambda)$$

1. What is the purpose of the study? 2. What are the research questions? 3. What are the hypotheses?

8. Wichtige Punkte des Vortrags

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1960-1961

1. Identify the subject and predicate of each sentence.

1. Welche Arbeit ist in dieser Mitteilung zu leisten?

... ..
: Detachable

[Faint handwritten notes at the bottom of the page]

1. James Smith & John Smith, Merchants, 1812

17

11: 11111
 12: 11111

24th Oct 21

[illegible]

MODEL SECONDARY SCHOOL

KATHMANDU MODEL SECONDARY SCHOOL

1. *Chlorophyll a* (Chl a) is the primary photosynthetic pigment in most plants and algae. It is a green pigment that absorbs light energy in the blue and red regions of the visible spectrum. Chl a is essential for the light-dependent reactions of photosynthesis, where it converts light energy into chemical energy in the form of ATP and NADPH.

पूर्व वैमासिक परीक्षा-२०८२

क्षा : ११

दिनांक : १-३० मिनट

विषय : नेपाली

(००११)

पुर्णाङ्क : ३७

उत्तीर्णाङ्क : १५

प्रश्नहरू सकेसम्म आफ्नै भाषामा उत्तर दिनुपर्ने छ । दायाँ किनारामा दिइएको अङ्कले पुर्णाङ्क बताउँदछ ।

प्रश्नहरू अनिवार्य छन् ।

दिइएका व्यञ्जन वर्णहरूको उच्चारण स्थान र प्रागल्भ छुट्याई लेख्नुहोस् :

ख, त, प

दिइएको वाक्यलाई शुद्ध गरी पुनर्लेखन गर्नुहोस् :

३ वाक्यकालको दसैको स्मृति मेरो हृदयमा अफ सम्म ताने छ म पुर्वी नेपालको पहाडमा जन्मेको थिएँ । मेरो गाउँ भन्दा भन्दा दसैको यकल सस्कृतिमा जम्प्यो र त्यसमै रमाउँथ्यो ।

दिइएको अनुच्छेदमा रेखाङ्कित पदहरूको पदवर्ग पहिचान गरी लेख्नुहोस् :

मानिसको शोभा र सभ्यताको रक्षा गर्ने कुरा साज हो । त्यो साज मानिसमा छैन भने त्यस नाइगो मानिसमा कसको के लाग्छ ?

दिइएका वाक्यहरूलाई बहुवचनमा परिवर्तन गरी पुनर्लेखन गर्नुहोस् :

मैले गाउँमा वास्यकालको साधीलाई भेटेँ । मैले जीवन र जगतका बारेमा धेरै बेर कुरा गरेँ । साधीले आफ्नो प्रगतिको कथा सुनायो । म जित्छु परेँ ।

दुवै प्रश्नहरूको उत्तर दिनुहोस् :

(४+४=८)

क. दिइएका अनुच्छेद पढी सोधिएका प्रश्नहरूको उत्तर दिनुहोस् :

मानिस सासांरिक वस्तुहरूको अनुभव पाँच ज्ञानेन्द्रियबाट गर्दछ । उसका मुख्य नाडीहरू मस्तिष्कसँग मिल्नेका हुन्छन् । आँखा, नाक, कान, जिह्वो र छाला ज्ञानेन्द्रियहरू हुन् । मानिसले ससारको अनुभव धिनै पाँच इन्द्रियहरूले गर्दछन् । यी इन्द्रियहरूलाई हामीले सधैं स्वच्छ र सरल राख्नु पर्दछ । सामर्थ्यभन्दा अधिक काम तिनले इन्द्रियहरूमाथि आघात पुग्दछ । फलस्वरूप मस्तिष्कको क्षमता कमजोर हुन गई मानिसको स्मरणशक्ति कम हुन जान्छ । मस्तिष्कलाई सबल र स्वस्थ बनाउन शुद्ध रगतको आवश्यकता पर्दछ । दूला दूला महात्मा एवम् बुद्धिमान् पुरुष बीसको रक्षा गर्दा गर्दै पनि निर्बल प्रतीत हुन्छन् । यसको कारण के हो भने तिनले मस्तिष्कबाट धेरै काम लिनु पर्दछ, जसबाट रक्तादि पोषक भाग विचार शक्तिको साथ अधिक खर्च हुन्छ । रात दिनको पढाइको

कारणले विद्यार्थीको शारीरिक बल पनि लुट्न पुग्छ हुन जान्छ । मस्तिष्कलाई बलवान् बनाउनको लागि सवगुणी खाद्य पदार्थको सेवन गर्नुपर्छ । दुध, घिउ, चामल, गहुँ, केरा, अदुसुर, स्वाद आदि पदार्थमा सवगुण रहेको हुन्छ । तमीगुणी पदार्थले शरीरलाई भेटो बनाउँछ भने रजोगुणी पदार्थले शक्तिनाई शीघ्र व्यय गरिदिन्छ । त्यसैले यस्ता पदार्थको प्रयोग कम गर्दै जानुपर्छ ।

क) मानिसले ससारको अनुभव केबाट गर्छ ?

ख) मानिसको मस्तिष्कलाई सबल राख्ने उपायहरू के-के हुन् ?

ग) मस्तिष्कको क्षमता कमजोर हुनाका कारणहरू के-के हुन् ?

घ) शीघ्र र सामर्थ्य शक्तिको जर्ब लेख्नुहोस् ।

ख) दिइएको कवितामा पढी सोधिएका वस्तुगत प्रश्नहरूको उत्तर लेख्नुहोस् :

यहाँ पनि

म जस्तै जर्को नदी रहेछ ।

दोभानमा भेट भएपछि

दुई फरक बाटाबाट आएका नदीहरू

आफू जस्तै जर्को नदी देखेर छक्क परे ।

दुवैको यात्रा उस्तै थियो

सङ्घर्षले भरिएको बाटो

उस्तै थियो हृदयको सङ्गीत उस्तै थियो

प्रेमको सपना उस्तै थियो ।

कसलाई जित्नु छ र जब ?

दोभानमा भेट भएपछि

बाँकी यात्रा पूरा गर्न

उनीहरू सँगसँगै मिलेर बग्न थाले । - पिप्पल ढकाल

प्रश्नहरू :

(क) नदीहरू किन छक्क परे ?

(अ) आफू जस्तै खोला देखेर

(आ) थिचिच स्वस्वरूपका मान्छे देखेर

(इ) एउटा माछाले जर्को माछालाई निलेको देखेर

(ई) आफू जस्तै जर्को नदी देखेर

(ख) दुई नदीका थिच के के कुरामा सभानता छ ?

(अ) सङ्घर्ष, सङ्गीत र व्यथा

(आ) सङ्घर्ष, सङ्गीत र सपना

(इ) सुख, समृद्धि र सपना

(ई) सुख, समृद्धि र स्वभाव

(ग) दुई नदी किन सँगसँगै बग्न थाले ?

(अ) सुखदुःखका भावना साटोसाट गर्न

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
 PRE - TERM EXAMINATION 2082

Class: XI 'Science'
 Date: 1.5 hrs.

DAY SHIFT 'SET-B'
 Subject: Physics [1011]

F.M: 37
 P.M: 15

Students are required to give their answers in their own words as far as practicable. The figures in margin indicate full marks.

Attempts all the questions.

GROUP - 'A' [6x1=6]

Multiple Choice Questions:

The equation of a wave is given by, $y = A \sin \left[\omega \left(\frac{x}{v} - k \right) \right]$, when ω is angular velocity, x is position and v is linear velocity. The dimensional formula for k is:

- a) $[LT]$ b) $[T]$ c) $[T^{-1}]$ d) $[T^2]$

Identify the correct statement:

- a) Velocity, displacement, acceleration, force are vectors
 b) Weight, mass, torque and energy are scalars
 c) Angular velocity, momentum and electric current are scalars
 d) A vector quantity is one which has magnitude only whereas scalar quantity is one which has both magnitude and direction.

When 2×10^{18} electrons are added to a neutral metal sphere, the charge on the sphere becomes:

- a) $-1.6 \mu C$ b) $1.6 \mu C$ c) $3.2 \mu C$ d) $-3.2 \mu C$

A soap bubble is given a negative charge, then its radius:

- a) Decreases b) Increases c) Remains unchanged
 d) Nothing can be predicted as information is insufficient.

The force between two charges 0.06 m apart is 5 N. If each charge is moved towards the other by 0.01 m, then the force between them will be

- a) 7.20 N b) 11.25 N c) 22.50 N d) 45.0 N

When a positively charged conductor is connected to earth:

- a) No charge flow occurs
 b) Electrons flow from conductor to earth
 c) Protons flow from conductor to earth
 d) Electrons flow from earth to conductor

GROUP - 'B' [5x3=15]

Answer Questions:

What are the basic rules of dimensional analysis?

[1]

b) Can we tell the unit of a physical quantity from its dimensions? What about dimensions from the unit of a physical quantity? [1]

c) Check the correctness of the physical equations: (i) $t = 2\pi \sqrt{\frac{l}{g}}$; t is time period, l is length and g is acceleration due to gravity (ii) $T^2 = \frac{4\pi^2 a^3}{GM}$; T is time period, G is gravitational constant, M is mass and a is radius of the orbit. [3]

8. a) What is meant by resultant vector. [1]

b) Two vectors acting on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangular representation? State and explain the law involved. [3]

c) Under what condition is the magnitude of resultant vector is zero vector? [1]

9. a) Explain about Action of point with a demonstration? [2]

b) What do you mean by electrostatic induction? How will you give permanent positive charge to a conductor by the method of induction? Explain in detail. [3]

GROUP - 'C' [2x8=16]

Long Answer Questions

10. a) What do you mean by dimension of a physical quantity? What is the difference between dimensions and unit of a physical quantity? [2]

b) A body moving through air at a speed v experiences a retarding force F given by the equation $F = kApv^x$, where A is the surface area of the body, p is the density of air and k is dimensionless constant. Deduce the value of x . [2]

c) Convert 60 J/s into a new system of unit which has 10gm, 10 cm and 10 second as fundamental units of mass, length and time. [2]

d) Is it possible to derive all the physical equations by the method of dimensional method? Explain. [2]

11. a) Write the formula for Coulomb's force in vector form? What is the significance while expressing in vector form? [2]

b) What is electrical permittivity? Write its value and unit for vacuum in SI and cgs system of unit. [2]

c) If a body gives out 10^9 electrons per second, how much time is required to get a total charge of 1 C from it? What conclusion would you draw from the result obtained? [2]

d) Two positive point charges are at a separation of 0.3 m repel each other with a force of 1.6 N. If the sum of two charges be 10 microcoulomb, what are their separate values? [2]

Best of Luck

Students are required to give their answers in their own words as far as practicable. The figures in margin indicate full marks.

Time: 1.5 hrs.

Section: 'X' Science

The correct formula for the displacement of a particle executing periodic motion is (v, T, a, y and t represent velocity, time period, amplitude, displacement and time respectively)

- a) $y = a \sin \frac{2\pi}{T} t$ b) $y = a \sin vt$ c) $y = a \sin \frac{a}{2t}$ d) $y = a \sin \frac{a}{2t}$

A vector remains unchanged on:

- a) Rotating it through some angle
 b) Taking its dot product with a unit vector
 c) Shifting it parallel to itself
 d) Taking its cross product with a unit vector

When 2×10^{14} electrons are removed from a neutral metal sphere, the charge on the sphere becomes:

- a) $-16 \mu C$ b) $16 \mu C$ c) $32 \mu C$ d) $-32 \mu C$

roughly near, the force q_1 exerted on q_2 :

- a) Increases b) Increases c) Remains unchanged d) Decreases
 e) Increases f) Same sign as q_1 and decreases if q_2 is of the opposite sign.

The dielectric constant for a metal is:

- a) 1 b) 0 c) Greater than 1 d) infinity

You are travelling in a car during a thunderstorm. In order to protect yourself from lightning would you prefer to:

- a) Remain in the car
 b) Take shelter under a tree
 c) Touch the nearest electric pole
 d) Touch the ground

Do all the physical quantities have dimensions? Justify your answer. [1]

In CGS system, the value of Stefan's constant σ is $5.67 \times 10^{-8} \text{ erg } s^{-1} \text{ cm}^{-2} \text{ } ^\circ K^{-4}$. Find its value in international system of unit. [1]

GROUP - 'B' [5X3=15]

Answer Questions:

8. a) What is the necessary condition for adding of vectors.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
 c) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 d) Explain in detail. [1]
 e) permanent negative charge to a conductor by the method of induction? How will you give [1]

9. a) Repulsion is the surest test of electrostatic induction. Explain? [1]
 b) What do you mean by electrostatic induction? [1]
 c) Repulsion is the surest test of electrostatic induction. Explain? [1]
 d) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 e) Explain in detail. [1]

10. a) A physical equation is dimensionally correct. Is it necessary that the equation is physically correct?
 b) Taking force (F), length (L), and current (A) as fundamental quantities, find the dimensional formula for permeability μ_0 . [2]
 c) The gravitational constant is $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$. Express G in terms of length in astronomical units (1 AU = $1.496 \times 10^{11} \text{ m}$). Mass in solar masses (1 $M_\odot = 1.989 \times 10^{30} \text{ kg}$). Time in years by dimensional method. [3]
 d) While deriving the relationship between physical quantities by dimensional analysis, dimensional constants enter into a relationship. Can you find its magnitude by the method of dimensions? If not, how will you find? [1]

11. a) State and explain Coulomb's law of force between two stationary charges. Also, write the formula for Coulomb's force in vector form. [2+1+1]
 b) Two particles A and B having charges +4 nC and -1 nC respectively are held fixed with a separation of 0.20 m. Where should a third charged particle be placed so that it does not experience a net electrostatic force. [2]
 c) Two identical spheres having unequal opposite charges are placed at a distance 30 cm apart. After touching them mutually, they are again separated at the same distance apart. Now they repel each other with a force of 0.025 N. Calculate the final charge on each of them. [2]

GROUP - 'C' [2X8=16]

12. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
 c) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 d) Explain in detail. [1]
 e) permanent negative charge to a conductor by the method of induction? How will you give [1]

13. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
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14. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
 c) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 d) Explain in detail. [1]
 e) permanent negative charge to a conductor by the method of induction? How will you give [1]

15. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
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16. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
 c) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 d) Explain in detail. [1]
 e) permanent negative charge to a conductor by the method of induction? How will you give [1]

17. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
 c) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 d) Explain in detail. [1]
 e) permanent negative charge to a conductor by the method of induction? How will you give [1]

18. a) Check the correctness of the physical equations: i) $P = \frac{1}{2} \rho v^2 + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h is height. ii) $T^2 = \frac{4\pi^2 R^3}{GM}$; T is time period, G is gravitational constant, M is mass and R is radius of the orbit.
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangle representation? State and explain the law involved.
 c) Under what condition is the magnitude of resultant force is zero? Newton's [1]
 d) Explain in detail. [1]
 e) permanent negative charge to a conductor by the method of induction? How will you give [1]